

Optimizing Grazing Management for Profit and Ecosystem Services: An Integrated Biogeochemical-Economic Model

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Abstract. Grazing management jointly shapes livestock returns and ecosystem services, yet quantitative guidance on balancing them remains scarce. We develop an integrated daily biogeochemical-economic framework that links soil carbon dynamics and forage growth with rotation decisions, animal selective grazing, animal growth, herd management, and farm profit. To make this coupled system computationally tractable, we develop a recurrent surrogate of the Microbial Efficiency-Matrix Stabilization model that provides an approximately 181-fold speedup. Results show that calendar-based rotation's private value depends on stocking pressure. The maximum ten-year net present value is \$200,046 at 2.0 livestock units per hectare, two paddocks, and a 20-day grazing period, exceeding continuous grazing at the same stocking rate. Profit and SOC rankings nevertheless diverge. Calendar-based rotation also reduces patch-use imbalance, highlighting the redistribution of grazing pressure as one mechanism connecting grazing management to economic and environmental outcomes.

1. Introduction

Grazing lands support livestock production while also providing soil-carbon storage, water regulation, habitat, and rural livelihoods (Hruska et al. 2017; Briske et al. 2023). Management therefore has consequences beyond current livestock revenue: stocking pressure, paddock design, and movement timing jointly determine forage use, feed requirements, and environmental outcomes (Teague et al. 2016). Yet producers and conservation programs still lack clear quantitative guidance about when rotational grazing pays and how privately attractive management compares with management that protects ecosystem services.

Two features make this problem difficult to evaluate. First, soil organic carbon (SOC) is productive natural capital. Grazing management can gradually change SOC, and those changes can affect future forage quantity and quality and, in turn, purchased-feed requirements. Because SOC responds slowly, short experiments may observe current forage and livestock outcomes without capturing this longer-run response. Models that track forage without carrying SOC as a state likewise miss the feedback from current grazing to future forage productivity and feed costs. This mismatch in time scales is one reason the evidence on rotation is mixed (Briske et al. 2008; Wang, Teague, and Park 2016; Wang et al. 2018). Second, cattle graze selectively, and rotation outcomes depend on the herd, the timing rule, the spatial distribution of grazing, and the condition of the underlying resource. Cattle concentrate use on preferred parts of a pasture and leave other areas lightly used (Gersie, Augustine, and Derner 2019; Adler, Raff, and Lauenroth 2001); a homogeneous representative-paddock model cannot show how fencing and calendar movement redistribute that pressure. A useful economic model must therefore connect annual farm decisions to daily, spatially explicit animal–forage–soil dynamics while retaining SOC as a slow state variable.

Existing research supplies the components of that framework, although no study connects them completely. Soil-as-capital models define the shadow value of SOC primarily for cropland (Burt 1981; McConnell 1983; Berazneva et al. 2019). Grazing bioeconomic models examine stocking or rotation but generally omit detailed SOC dynamics, while process-based ecosystem models can represent those dynamics but are costly to evaluate repeatedly and leave the producer’s decision problem unspecified. Recent surrogate methods can accelerate process models (Reichstein et al. 2019; Liu et al. 2024), but speed alone is insufficient for management analysis: the farm model must translate producer choices such as herd target, paddock count, and grazing-period length into the daily local grazing inputs required by the surrogate. These gaps leave three questions: how should the private value of SOC enter grazing decisions; when does calendar-based rotation outperform continuous grazing; and how do the profit-maximizing and SOC-protecting management strategies differ?

We answer these questions with an integrated biogeochemical-economic model of a cow-calf farm. The theory extends the cropland soil-capital framework to grazing and defines the private shadow value of SOC through forage productivity and feed-cost savings. Conditional on installed paddock count, the producer chooses a nominal herd target and a calendar grazing-period length; scheduled rest is implied by those choices. Herd composition and animal demand translate the annual management choices into daily forage demand, while calendar eligibility and selective movement determine the local patch that is grazed. A numerical model then couples this farm-management system to the Microbial Efficiency-Matrix Stabilization (MEMS) process model (Zhang et al. 2024; Santos et al. 2024) at a representative 60-hectare Michigan site. Because repeated MEMS evaluation is computationally prohibitive, a management-responsive recurrent surrogate advances the complete daily biophysical state in the 663-cell management experiment.

Our main contribution is a unified decision framework that links SOC as productive capital to the choices and daily processes of a grazing farm. The framework extends soil-capital economics from cropping to grazing, where the private return to SOC operates through forage quantity and quality and the resulting feed bill. It treats installed paddock design, annual herd and timing choices, within-season movement, herd transitions, nutrition, and annual accounting consistently. In the calendar system, once paddock count is fixed, choosing how long cattle remain in each paddock also determines how long that paddock rests before the herd returns. A management-responsive recurrent surrogate makes the daily process model usable within the whole-farm simulation while leaving herd target, paddock design, grazing-period length, movement, and economic accounting in the farm model. The patch representation also measures how grazing is redistributed across heterogeneous locations, one spatial mechanism through which paddock design and timing affect forage use and, through the coupled biogeochemical and economic system, feed costs, profit, and SOC.

The results show that the privately preferred paddock design and grazing schedule depend on stocking pressure. The surrogate achieves a mean weighted absolute percentage error of 6.44 percent across eight management-relevant outputs over eight ten-year recursive trajectories, with no physical-validity violations, and reduces the runtime of a ten-year adaptive-management scenario from approximately one hour to 20 seconds. In the management sweep, the highest private net present value is \$200,046 at 2.0 livestock units per hectare, two paddocks, and a 20-day grazing period. That management configuration outperforms continuous grazing at the same stocking rate by 8.85 percent, largely through a lower grazing-season deficit-feed charge. The advantage of calendar-based rotation grows with stocking pressure, while private profit and SOC rank the management configurations differently. Modeled SOC declines in every cell relative to the baseline generated by an ungrazed MEMS history from 1926 to 2025. The least loss occurs at 0.8

livestock units per hectare, twelve paddocks, and a six-day grazing period, a different configuration from the private-profit maximum. Calendar-based rotation also greatly reduces patch-use imbalance, documenting this spatial channel alongside the other biophysical and economic mechanisms represented in the integrated model.

The scope of these scenario comparisons is bounded by one site, one forcing path, prescribed animal growth, and a discrete set of fixed management configurations. SOC is reported as an environmental outcome and carries no payment in the private objective. The findings therefore support conditional management guidance. The framework also establishes the economic and computational foundation for evaluating how climate, markets, and ecosystem-service incentives could change preferred grazing management.

The remainder of the paper proceeds as follows. Section 2 develops the SOC motivation and positions the paper in the soil-capital, grazing-bioeconomics, spatial-grazing, surrogate-modeling, and payment-design literatures. Section 3 presents the bioeconomic framework. Section 4 describes the Michigan application, MEMS surrogate, coupled farm simulation, parameterization, and experimental design. Section 5 reports surrogate validation and computational performance, Section 6 presents the management-scenario results, and Section 7 concludes with implications and limitations.

2. Soil carbon in grazing-management economics

Soil organic carbon is the slow-moving stock at the center of the grazing system. It changes little from one year to the next, yet over a decade it governs both the forage a pasture can grow (Lal 2004; Stanley et al. 2024) and the climate benefit a grazing operation can claim (Conant, Paustian, and Elliott 2001; Teague et al. 2016). What sets the grazing case apart from the cropland settings where soil-as-capital models were first built is the mechanism from carbon to net value. In cropping, soil organic carbon raises crop yield and revenue through an agronomic response. In grazing the link occurs through SOC's impact on forage quantity and quality and, in the present accounting, on the stored-feed quantity implied by nonnegative daily class-specific pasture-energy deficits and the off-season feeding requirement. Since stored feed substitutes economically for pasture-supplied energy, the value of SOC depends on its ability to produce more forage within a season and reduce the annual feed bill.

Treating soil as a depletable capital stock whose value should enter farm-level decisions began with Burt (1981) and McConnell (1983), the latter deriving the shadow value λ of soil that at steady state satisfies $\lambda^{ss} = (\partial\pi/\partial\text{SOC})/(1 - \delta \partial\text{SOC}_{t+1}/\partial\text{SOC}_t)$. We follow Berazneva et al. (2019), who calibrated SOC dynamics from the Rothamsted Carbon Model and an econometric maize-yield function for smallholder Kenyan farms and solved the steady-state optimality conditions simultaneously for shadow values of \$95 to \$168 per Mg C. We share their state variable and solution strategy but differ in the production system: their

carbon-to-revenue link is the direct response of crop yield to SOC, whereas ours runs indirectly through forage quantity and quality and through the annual stored-feed bill that more abundant forage reduces.

Pairing a calibrated biophysical model with an estimated economic one, as in Be-
razneva et al. (2019), is an instance of the Econometric-Process Model (EPM) strategy of
Antle and Capalbo (2001), in which a biophysical model maps management to yields and
stocks and an economic model chooses management in response. Antle et al. (2003) used
this strategy to design carbon-sequestration contracts, and Antle et al. (2007) applied it
at a national scale to estimate agricultural carbon supply curves. Our coupling of MEMS
with a stocking-and-rotation optimization adapts this EPM strategy to a process-based
grassland model.

Grazing land (or grassland) soil carbon accumulates differently from cropland. It
enters through root turnover and the manure the herd returns to the pasture, and because
pasture is perennial and untilled it avoids the tillage that accelerates cropland carbon loss
(Stanley et al. 2024; Teague et al. 2016). That improved management can build the SOC
stock is well documented: Conant, Paustian, and Elliott (2001) synthesized 115 studies that
found a mean gain of 0.54 Mg C per hectare per year. We use this estimate as empirical
context for the single-site model. Lal (2004) ranked grazing among the few practices with
material climate-mitigation potential. In cropland settings, soil-health practices capitalize
into farmland prices (Chen et al. 2023, 2026) and build resilience to climate stress (Aglasan
et al. 2024), and Chen et al. (2025) solve a cover-crop adoption problem with soil fertility
as the state variable, the closest cropland analog to ours and a method we draw on.

A parallel grazing-bioeconomics literature supplies the dynamic structure. Grazing
systems can exhibit nonmonotonic responses: Noy-Meir (1975) (generalized by May (1977))
showed that moderate stocking admits both a productive and a degraded stable state
separated by a threshold across which a small stocking change can tip the system into
collapse. Huffaker and Cooper (1995) separated rangeland into a fast forage process and
a slow successional one and formalized the split as a slow-manifold decomposition, a
structure we keep with SOC as the slow stock, while Huffaker and Wilen (1991) cast within-
season forage decline as a fishery-style control problem. Whether the slow stock matters
is what divides long-horizon from short-horizon analyses. Torell, Lyon, and Godfrey (1991)
found the two nearly identical when only forage was tracked, precisely because their
model held no slow stock, and Ritten et al. (2010) extended that deterministic model to
stochastic dynamic programming under variable weather. Ritten, Bastian, and Rashford
(2012) found carbon payments at prevailing prices well below the management costs they
were meant to cover. We therefore track production and SOC within one dynamic system
and exclude standalone carbon revenue from the private objective. The value of rotation
itself is contested: Wang, Teague, and Park (2016); Wang et al. (2018) found that multi-

paddock rotation raises ranch net present value and Derner et al. (2021) and Bosch et al. (2008) report ranch-scale gains, while Briske et al. (2008) found no advantage in controlled plot-scale experiments. Those experiments are too short to resolve the decade-scale SOC feedbacks our process-based simulation is built to capture, and we revisit that debate below.

Grazing further differs from cropping because cattle use space unevenly within a farm. By GPS tracking, Gersie, Augustine, and Derner (2019) found that 4 to 16% of a pasture went ungrazed while a small fraction received more than four times the average use, the pattern sharpening as forage quality differentiated through the season, and Adler, Raff, and Lauenroth (2001) showed that such preferential use reshapes vegetation into a mosaic of heavily used lawns and lightly used roughs that carries through to the soil. Zhang et al. (2026) embedded explicit livestock movement in MEMS to capture this within-pasture variation, the closest modeling predecessor to our patch-based representation. Our focus on within-farm heterogeneity complements the between-site heterogeneity Antle et al. (2003) used for contract design. A representative-paddock model cannot express this spatial pattern, so we represent the farm as patches grazed preferentially and let SOC diverge across them under management.

The optimization template comes from renewable-resource theory. Gordon (1954) showed that open access drives a stock below the social optimum, Schaefer (1954) supplied the logistic growth backbone, and Clark (1976) synthesized fishery economics into the modified golden rule that is the direct ancestor of our shadow-value formula, with Smith (1969) adding the effort dynamics that parallel herd expansion and contraction. Homans and Wilen (1997) showed that a season-length catch limit lets fishing capacity expand until rents dissipate, which maps onto practice-based grazing payments that may invite overstocking within a rotation. Our within-season problem chooses the calendar length of a grazing bout conditional on an installed paddock design; the resulting rest interval is nested inside the across-year Bellman equation for SOC. Its daily dynamics resemble the drawdown of fish stocks in Homans and Wilen (1997), with the farmer as the resource manager and with multiple spatially distinct drawdown periods per year.

Running process-based models requires significant computation time. Scaling this single-site analysis to many sites and repeatedly evaluating management therefore motivates a surrogate trained on process-model output. Reichstein et al. (2019) argued that hybrid mechanistic-and-learning models suit Earth-system problems, and Liu et al. (2024) realized this for cropland carbon, training networks on the *ecosys* model to $R^2 > 0.9$. Such surrogates can be trained from process simulations (Lu and Ricciuto 2019) and serve as a bridge from mechanistic models to economic integrated assessment (Blanc 2017; Franke et al. 2020). On the economic side, Shang et al. (2024) built a deep-learning surrogate of a farm optimization model. Our structured recurrent surrogate emulates the complete

daily MEMS state and accepts two management inputs directly: local LSU per hectare and optional on-pasture supplement. This design lets the same farm-management and animal modules evaluate hundreds of management configurations without explicitly constructing MEMS's internal action fluxes.

The framework is also intended to support future comparisons of carbon offsets, practice-based payments, and price premiums tied to qualifying production practices. This application connects to the literature on payment design for ecosystem services. Antle et al. (2003) showed per-tonne (outcome-based) carbon contracts to be roughly 5× more efficient than per-hectare (practice-based) contracts in the dryland Northern Great Plains, with the gap driven by spatial heterogeneity in sequestration potential, both across sites and within a single farm. Rouhi Rad et al. (2025) obtain an analogous result for groundwater conservation, finding that incremental abatement subsidies dominate retirement programs. Engel, Pagiola, and Wunder (2008) laid out the three dimensions of payment for environmental services design: conditionality on outcomes, additionality of conservation, and targeting of payments to high-impact sites. Melkonyan and Taylor (2013) compared input regulation, cost-sharing, and performance regulation for public range-lands and found performance regulation theoretically efficient but operationally costly. The additionality problem received its sharpest formulation in Ferraro and Pattanayak (2006) (“money for nothing”) and was extended by Pates and Hendricks (2020) to dynamic settings where technology diffusion erodes additionality over time. Wongpiyabovorn, Plastina, and Crespi (2023) documented the practical barriers to outcome-based payments (fixed certification costs of \$5,000 or more and ongoing verification expenses) that may exclude small cow-calf operations regardless of the theoretical efficiency advantage. Wang, Isengildina-Massa, and Stewart (2023) found that consumer premiums for grass-fed beef are driven mainly by health and taste; climate and carbon claims play a smaller role. This preference pattern complicates the use of market premiums as ecosystem-service payments.

These literatures have largely developed separately. Bioeconomic SOC models focus on cropland, grazing models generally omit detailed SOC dynamics, and grazing-land biogeochemistry remains largely absent from existing surrogate applications to crop-process and economic models. Studies of carbon-payment design likewise concentrate on cropland. One contribution of this paper is to integrate these strands in a coupled private-profit management framework that treats SOC as productive natural capital while also reporting it as a separate environmental outcome.

3. Bioeconomic framework for grazing management

The framework runs from the whole farm to the local patch and back. At the whole-farm level, the producer selects a nominal annual herd target and a calendar grazing

rule conditional on installed infrastructure. The herd controller converts that target into realized cow, heifer, and calf counts; the animal-demand schedule converts those counts into a daily demand-equivalent herd input. The calendar restricts the herd to a management paddock, animal selectivity determines the patch actually used, and the resulting local actions—forage removal, trampling, and animal returns—drive daily forage and soil transitions. At weaning and the year boundary, the herd-composition transition promotes animals between classes, records sales and replacements, and determines the next year’s herd. Aggregating the daily transitions and herd outcomes yields annual SOC, stored-feed cost, and profit for the producer’s dynamic problem.

3.1. Whole-farm management and dynamic decision problem

Years are indexed by t and calendar days within a year by d . Let $\mathcal{C} = \{\text{cow, heifer, calf}\}$ index animal classes. Let $\mathbf{X}_t$ collect the complete farm biophysical state at the start of year t , including the spatial distributions of vegetation, soil water and temperature, nutrients, surface organic matter, and SOC; $X_{k,t,d}$ denotes the vector state of patch k at the end of day d . Let $\mathbf{n}_t = (n_{c,t})_{c \in \mathcal{C}}$ collect opening animal counts. Finally, let $W_t = (w_{t,d})_d$ denote the year’s exogenous daily forcing path, including weather, atmospheric carbon dioxide, and calendar-indexed inputs. Farm area A , the grazing-season start and end dates d_{on} and d_{off} , and the herd-demographic parameters are fixed features of the production setting. The producer’s annual operating controls are the nominal whole-farm herd target H_t , measured in 500-kilogram reference LSU, and, for rotational designs, the calendar grazing-period length gp_t , measured in days spent in each management paddock before advancing. Because area is fixed, nominal stocking rate is $\text{SR}_t = H_t/A$. This nominal target differs from the demand-equivalent daily MEMS input defined below. When $N = 1$, the timing control is inactive and the maximization below is only over H_t .

The number of management paddocks N is a discrete, long-lived design choice selected from feasible set $\mathcal{N}$. Changing N requires fencing, which precludes costless annual adjustment. The operating problem is therefore solved conditional on installed N , fence investment is annualized in the farm account, and designs are compared separately. The exogenous forcing affects forage and soil transitions and, through the stored-feed requirement, current profit. Let $\mathcal{W}_{t:\infty} = (W_t, W_{t+1}, \dots)$ be a specified forcing sequence and let $\delta \in (0, 1)$ be the annual discount factor. Conditional on that sequence, the producer’s

time-dependent operating value is

$$(1) \quad V_{N,t}(\mathbf{X}_t, \mathbf{n}_t; \mathcal{W}_{t:\infty}) = \max_{H_t, \text{gp}_t} \left\{ \begin{aligned} &\pi_N(\mathbf{X}_t, \mathbf{n}_t, H_t, \text{gp}_t; W_t) \\ &+ \delta V_{N,t+1} \left(G_N(\mathbf{X}_t, \mathbf{n}_t, H_t, \text{gp}_t; W_t), \right. \\ &\quad \left. \mathcal{H}(\mathbf{n}_t, H_t); \mathcal{W}_{t+1:\infty} \right) \end{aligned} \right\}.$$

The preferred installed design is $N^* \in \arg \max_{N \in \mathcal{N}} V_{N,0}(\mathbf{X}_0, \mathbf{n}_0; \mathcal{W}_{0:\infty})$, with annualized fencing costs included in π_N . The function G_N advances the biophysical state and $\mathcal{H}$ advances herd composition. This formulation evaluates management conditional on a deterministic forcing scenario, exactly as in the numerical replay below. Under uncertain forcing, the same equation is written with an expectation conditional on the producer's information when H_t and gp_t are chosen. The following subsections unpack the mechanisms summarized by G_N , $\mathcal{H}$, and π_N .

3.2. Herd composition and transition model

The nominal herd target is expressed through class shares. These shares construct target class counts; observed opening proportions may differ, and the demographic controller does not reimpose the shares each day or year. Let α_c be the nominal share of class $c \in \mathcal{C}$, let ℓ_c be its representative LSU coefficient, and let $\bar{\ell} = \sum_{c \in \mathcal{C}} \alpha_c \ell_c$. The nominal class target associated with H_t is

$$(2) \quad \tilde{n}_{c,t}(H_t) = \alpha_c \frac{H_t}{\bar{\ell}}.$$

We use $\alpha = (0.50, 0.10, 0.40)$ and $\ell = (1.00, 0.70, 0.20)$, ordered as cows, replacement heifers, and the annual calf cohort. The vector $\tilde{\mathbf{n}}_t$ gives the nominal management target. Realized herd composition evolves through the demographic controller. At numerical initialization, the nominal cow target is translated into attrition-consistent opening cow and heifer counts, while the calf cohort remains pending until the calving date. Within year t , $n_{c,t,d}$ denotes the realized count of class c on day d after any scheduled demographic event.

The demographic balance for each class can be written generally as

$$(3) \quad n_{c,t+1} = (1 - \mu_c) n_{c,t}^{\text{end}} + b_{c,t} + p_{c,t}^{\text{buy}} - v_{c,t}^{\text{sale}} - q_{c,t}^{\text{cull}} + \sum_{c' \neq c} \tau_{c' \rightarrow c,t} - \sum_{c' \neq c} \tau_{c \rightarrow c',t},$$

Here $n_{c,t}^{\text{end}}$ is the class count immediately before year-boundary flows, μ_c is mortality, b denotes births, p^{buy} purchases, v^{sale} sales, q^{cull} culls, and τ promotions between classes.

In the implemented three-class rule, pending calves enter the live class at calving. At weaning, surviving calves are split by sex; male calves and female calves not needed for replacement are sold, existing heifers are promoted to cows, and retained female calves become the new heifer cohort. Replacement heifers are purchased only if retained females cannot fill the cohort needed to return the herd toward its target. Cow and heifer mortality and cow culling are applied at the year boundary, and the surviving cow herd and conception rule determine the next pending calf cohort. These fixed rules define $\mathcal{H}$, supply sales and replacement flows to annual accounting, and pass $\mathbf{n}_{t+1}$ to the next annual cycle. The same vector of realized class counts drives animal demand and farm accounting. All reported management scenarios use the same demographic rates and event dates. Although equation (1) permits annual retargeting of H_t , the reported numerical comparison holds $H_t \equiv H$ within each ten-year management scenario.

3.3. Calendar movement rule

Let the farm contain K biophysical patches indexed by k , with areas A_k satisfying $\sum_k A_k = A$. The installed design partitions those patches into N non-overlapping management paddocks,

$$\mathcal{G}(N) = \{\mathcal{P}_1, \dots, \mathcal{P}_N\}, \quad \bigcup_{j=1}^N \mathcal{P}_j = \{1, \dots, K\}.$$

The biophysical patches are the units at which local state is simulated; management paddocks are groups of patches used by the farm calendar.

The fixed grazing season has length $L_g = d_{\text{off}} - d_{\text{on}} + 1$. Under equal-duration sequential rotation, the full calendar cycle and the scheduled rest interval of a management paddock are

$$(4) \quad c_t = N \text{gp}_t, \quad r_t = (N - 1) \text{gp}_t.$$

The producer's primitive timing decision is grazing-period length. Once N and gp_t are set, the calendar determines rest. Farm area and paddock count alone are insufficient to determine a duration. Grazing-season dates remain fixed throughout the analysis.

Let $j_{t,0}$ denote the first management paddock used in year t . During the grazing season, the scheduler holds the herd in one management paddock for gp_t days and then advances sequentially to the next. The first paddock may rotate across years to distribute incomplete end-of-season cycles, but residual forage is an outcome and never triggers a management-paddock advance. When $N = 1$, the calendar places no within-farm restriction on the herd. Appendix A.1 gives the exact day-index rule and boundary bookkeeping.

3.4. Animal demand schedule

Given realized daily class counts and prescribed weights, the animal demand schedule calculates the energy and dry matter required to support the prescribed growth paths. For each animal class c , the weight trajectory is inverted through the nutrition equations to obtain $D_{c,t,d}^{\text{req}}$, the dry-matter intake per head needed to support that trajectory. Total herd requirement is

$$D_{t,d}^{\text{req}} = \sum_{c \in \mathcal{C}} n_{c,t,d} D_{c,t,d}^{\text{req}}.$$

The MEMS grazing interface reconciles dry-matter demand to its LSU control using a 500-kilogram reference animal, a daily intake capacity $\rho_D = 0.03$ of reference mass, and a calibration factor $\chi_{\text{MEMS}} = 0.86$. Thus one demand-equivalent LSU represents $\kappa_{\text{MEMS}} = m^{\text{ref}} \rho_D \chi_{\text{MEMS}} = (500 \text{ kg})(0.03)(0.86) = 12.9 \text{ kg DM day}^{-1}$, and the daily herd input is

$$(5) \quad L_{t,d} = \frac{D_{t,d}^{\text{req}}}{\kappa_{\text{MEMS}}}.$$

The annual target H_t therefore represents intended whole-farm herd size, whereas $L_{t,d}$ is the demand-equivalent herd that drives the daily biophysical model. Conditional on realized class counts, the prescribed weight path determines $D_{t,d}^{\text{req}}$ and hence $L_{t,d}$; realized forage leaves the prescribed weight gain unchanged in the reported scenarios.

3.5. Animal selective-grazing model

Within the active management paddock, cattle may use patches selectively. Let $\mathcal{E}_{t,d} = \mathcal{P}_{j_{t,d}}$ be the eligible set and let $\Omega(\cdot)$ be the grazing value of a patch as a function of its preceding state, the herd's preceding location, and fixed spatial attributes. The occupied patch is

$$(6) \quad k_{t,d} = \arg \max_{k \in \mathcal{E}_{t,d}} \Omega(X_{k,t,d-1}, k_{t,d-1}, k).$$

This abstract selection rule is sufficient for the theory; Section 4.5 specifies the numerical score. Under continuous grazing, all K patches remain eligible throughout the grazing season, and animal selective grazing determines the occupied patch through equation (6). Spatial use can therefore remain uneven under continuous grazing, which is represented numerically by $N = 1$. Under calendar-based rotation, $N > 1$ and the active management paddock restricts the eligible set. When $N = K$, every management paddock contains one patch and the calendar fully determines location.

3.6. Local grazing-pressure mapping

The annual management choices produce daily local grazing pressure

$$(7) \quad u_{k,t,d} = \mathbf{1}\{d_{\text{on}} \leq d \leq d_{\text{off}}\} \mathbf{1}\{k = k_{t,d}\} \frac{L_{t,d}}{A_k},$$

measured in LSU per hectare. The occupied patch receives the herd and every other patch receives zero.

3.7. Biophysical transition model

Stored-feed accounting is separate from the pasture–soil transition. Let $s_{k,t,d}$ denote an explicitly applied local supplement dry-matter flux in grams per square meter per day. The baseline sets $s_{k,t,d} = 0$ while still charging for S_t^{store} from equation (11). A positive on-pasture feed scenario would require a separate causal supplement-placement rule; Appendix B.1 gives that unused input mapping.

Local grazing pressure enters the transition as a management input and carries no carbon or nutrient flux directly. Let

$$(8) \quad \mathbf{a}_{k,t,d} = \mathcal{A}(X_{k,t,d-1}, u_{k,t,d}, s_{k,t,d})$$

denote the causal action map from the preceding patch state and current management inputs to realized forage removal, trampling, fecal carbon and nitrogen, urine carbon and nitrogen, and on-pasture supplement. Realized removal is capped by forage available at the start of the day, so all action components are known before the patch state is advanced.

Let z_k denote fixed site characteristics and $w_{t,d}$ the day's exogenous forcing and calendar context. The generic local transition is

$$(9) \quad X_{k,t,d} = F(X_{k,t,d-1}, z_k, w_{t,d}, \mathbf{a}_{k,t,d}),$$

where the arguments are all predetermined at the transition boundary. This representation deliberately leaves the transition technology unspecified. Section 4.3 describes its process-model and surrogate implementations.

3.8. Annual farm accounting

Feed account. The energy calculation reconciles the economic account with prescribed animal growth. For each animal class c and grazing day d , it compares the energy required by the prescribed growth path, $E_{c,t,d}^{\text{bench}}$, with the energy delivered by the class's body-size-weighted share of simulated pasture intake and quality, $E_{c,t,d}^{\text{past}}$. The nonnegative daily

grazing-season feed requirement is

$$(10) \quad S_t^g = \sum_{d=d_{\text{on}}}^{d_{\text{off}}} \sum_{c \in \mathcal{C}} \max \left\{ 0, \frac{E_{c,t,d}^{\text{bench}} - E_{c,t,d}^{\text{past}}}{e_s} \right\},$$

where e_s is usable energy per kilogram of hay-equivalent dry matter. Let $S_t^{\text{off}} \geq 0$ be the separately calculated winter feed requirement. The annual quantity priced as purchased feed is

$$(11) \quad S_t^{\text{store}} = S_t^{\text{off}} + S_t^g.$$

Both streams are nonnegative and priced separately. Consequently, surplus pasture energy for one class or day cannot cancel a deficit for another class or day, and grazing-season surplus cannot offset winter feed already consumed. The reconciliation maintains the prescribed growth path through the economic account alone; the surrogate's on-pasture supplement action remains zero. Appendix B.1 defines the daily class-specific energy streams and units.

Farm state and profit accounts. Let $\mathbf{X}_t = (X_{1,t}, \dots, X_{K,t})$ collect the patch states at the start of year t , and let $\text{SOC}(X_{k,t})$ extract total SOC from patch k . Farm-average SOC is

$$(12) \quad \text{SOC}_t = \sum_{k=1}^K \frac{A_k}{A} \text{SOC}(X_{k,t}).$$

Composing equation (9) over all days and patches under the calendar, selection, feed, and local-pressure rules defines the annual farm transition

$$(13) \quad \mathbf{X}_{t+1} = G_N(\mathbf{X}_t, \mathbf{n}_t, H_t, \text{gp}_t; W_t), \quad \text{SOC}_{t+1} = g_N(\mathbf{X}_t, \mathbf{n}_t, H_t, \text{gp}_t; W_t),$$

with $\mathbf{n}_{t+1} = \mathcal{H}(\mathbf{n}_t, H_t)$, where W_t is the annual exogenous forcing path. The complete transition depends on the spatial distribution of state and herd composition in addition to farm-average SOC. The scalar function g_N used below is therefore a reduced-form SOC projection of the full daily system, conditional on its other state components, the herd state, forcing, and the fixed feed-placement rule.

Annual profit conditional on installed design N is

$$(14) \quad \pi_N(\mathbf{X}_t, \mathbf{n}_t, H_t, \text{gp}_t; W_t) = R_{\text{calf}} + R_{\text{cull}} - C_{\text{feed}} - C_{\text{herd}} - C_{\text{min}} - C_{\text{lab}}(N, \text{gp}_t) - C_{\text{fence}}(N) - C_{\text{repl}},$$

with every term evaluated for year t under forcing path W_t . Calf and cull revenue follow the herd schedule. Feed cost C_{feed} prices the two nonnegative quantities in equation (11),

with winter and grazing-season feed charged separately. Both remain outside the baseline pasture transition. Herd nonfeed cost and mineral cost are recorded separately as C_{herd} and C_{min} . Rotation affects labor through scheduled management-paddock moves and fencing through installed N , with the durable fence cost converted to an annual equivalent. Higher SOC can raise private profit by increasing forage quantity and quality and thereby lowering C_{feed} . Terminal SOC and the carbon benchmark are reported as separate environmental criteria and are excluded from π_N .

The model distinguishes forage, SOC, myopic-profit, and forward-looking management benchmarks. Their formal definitions appear in Appendix A.2. A longer calendar grazing period simultaneously lengthens scheduled rest, changes movement frequency, and changes how animal selective grazing distributes use across patches. The resulting ranking is an empirical property of G_N and π_N , and the calendar identity alone leaves it indeterminate.

3.9. Dynamic optimization and SOC valuation

With the annual transition and profit components now defined, equation (1) gives the private-profit operating problem conditional on installed infrastructure. The outer comparison over N respects the discreteness and durability of paddock design and avoids treating fencing as a costless annual adjustment.

For a transparent scalar shadow-value expression, condition on the non-SOC components of $\mathbf{X}_t$, the herd state $\mathbf{n}_t$, and a stationary reference forcing environment $\bar{W}$, and evaluate both profit and the transition under that same environment. Suppressing these conditioned arguments gives the autonomous reduced operating problem

$$\tilde{V}_N(\text{SOC}_t) = \max_{H_t, \text{gp}_t} \{ \pi_N(\text{SOC}_t, H_t, \text{gp}_t) + \delta \tilde{V}_N(g_N(\text{SOC}_t, H_t, \text{gp}_t)) \}.$$

Let $\lambda_{t+1} = \tilde{V}'_N(\text{SOC}_{t+1})$ denote the private marginal continuation value of farm-average SOC. For a rotational design $N \geq 2$, an interior solution satisfies

$$(15) \quad \frac{\partial \pi_N}{\partial H_t} + \delta \lambda_{t+1} \frac{\partial g_N}{\partial H_t} = 0,$$

$$(16) \quad \frac{\partial \pi_N}{\partial \text{gp}_t} + \delta \lambda_{t+1} \frac{\partial g_N}{\partial \text{gp}_t} = 0.$$

The herd condition balances the current marginal return to another nominal LSU against its discounted SOC effect. The calendar condition balances the current feed, labor, and movement consequences of changing the grazing period against its discounted SOC effect. Holding the nominal herd target and installed N fixed, let $\text{gp}^m(H; N)$ denote a unique interior myopic timing optimum, so $\partial \pi_N / \partial \text{gp} = 0$. If $\lambda_{t+1} > 0$, $\partial g_N / \partial \text{gp} > 0$ there, and the

dynamic objective is strictly concave in gp over the feasible interval, the forward-looking timing choice satisfies $gp^d(H; N) > gp^m(H; N)$. By equation (4), scheduled rest is then also longer. Joint adjustment of the herd target and timing requires the full system in equations (15) and (16). The comparative result applies only to timing conditional on fixed H and N ; the forage, profit, and SOC optima may follow different rankings.

Applying the envelope theorem and imposing a stationary scalar equilibrium yields the private steady-state shadow value of SOC:

$$(17) \quad \lambda_N^{ss} = \frac{\partial \pi_N / \partial SOC}{1 - \delta \partial g_N / \partial SOC}.$$

It rises with the marginal feed-cost saving produced by SOC, the persistence of SOC, and the producer's patience. The scalar expression is exact for the autonomous reduced problem under $\bar{W}$. The numerical experiment uses a nonstationary finite forcing path: it replays historical weather while atmospheric carbon dioxide follows simulation year. The resulting paths support finite-horizon management comparison. Direct computation of the steady-state shadow value requires the stationary reference environment. The numerical model retains the complete patch-state vector and can evaluate how the reduction depends on the distribution of state across the farm.

For each installed N , Appendix A.4 gives four stationary equations for nominal herd target, grazing period, SOC, and the SOC shadow value. Eliminating λ_N^{ss} from its two operating conditions gives the operating-management isoquant

$$(18) \quad \frac{\partial \pi_N / \partial H}{\partial \pi_N / \partial gp} = \frac{\partial g_N / \partial H}{\partial g_N / \partial gp}.$$

It compares the two annual operating margins conditional on N . Installed paddock designs require a direct comparison of their values because the design choice is discrete.

3.10. Spatial aggregation

The farm-management system explicitly separates management paddocks from biophysical patches. Real cattle concentrate on preferred locations and leave others lightly used (Gersie, Augustine, and Derner 2019), and that preferential use reshapes vegetation and, through it, soil (Adler, Raff, and Lauenroth 2001). Equation (6) therefore lets patch states diverge even when all patches start with the same soil and weather. Farm-average SOC in equation (12) is an aggregation of those heterogeneous stocks. With a nonlinear daily transition, its shadow value may differ from the area-weighted average of patch-specific values. The reduced scalar problem provides the transparent shadow-value expression; the full transition G_N preserves the spatial distribution needed to evaluate it.

3.11. Theory-to-numerical mapping

The steady-state system (SS-1)–(SS-4) and the operating-management isoquant (18) are reduced-form implications conditional on an installed N and stationary reference forcing. The theory establishes the decision structure and a conditional comparative result: holding the nominal herd target fixed, a forward-looking producer chooses a longer calendar grazing period than the myopic producer when the marginal SOC effect of gp remains positive at the myopic optimum. The ranking of forage, profit, and SOC optima remains an empirical question, as does the magnitude of λ_N^{ss} . Both depend on the daily transition and profit functions under the chosen reference environment.

We give the framework empirical content by specifying each component of equations (8) and (9). The numerical animal model supplies total dry-matter requirement $D_{t,d}^{req}$, its demand-equivalent daily herd input $L_{t,d}$, class-specific benchmark energy, and the positive off-season hay requirement. The biophysical transition supplies realized pasture intake and quality, which are allocated across classes by metabolic body size. Equation (10) converts each class-day energy deficit to a nonnegative grazing-season feed quantity. In the single-site demonstration, each herd class follows a fixed growth path, so sale weights and per-animal revenue are held constant across management. Soil carbon’s private value is realized through the forage quantity and quality that reduce purchased feed.

The analytical and numerical controls coincide. Both condition on an installed paddock count N , use nominal herd target H and grazing-period length gp as the operating margins, hold grazing-season dates fixed, and derive scheduled management-paddock rest as $(N - 1)gp$. Both use selective patch use within the active paddock and map the resulting location and demand-equivalent herd $L_{t,d}$ into daily local LSU per hectare. The numerical experiment restricts $H_t \equiv H$, evaluates a discrete grid of fixed management configurations with the management-responsive surrogate, and conditions on one forcing replay. The Bellman problem remains a theoretical benchmark for future optimization. The numerical private objective is ten-year discounted profit; terminal SOC is reported separately. The purchased-feed account follows equations (10)–(11), and $s = 0$ keeps purchased feed outside the biophysical transition in every management cell.

4. Numerical models and experimental design

4.1. Study system, farm, and herd

We demonstrate the model at a single cow-calf site in Michigan. We assume a representative 60-hectare farm, corresponding in scale to the 50-to-99-head herd class reported for Michigan in the 2022 USDA Census of Agriculture (USDA-NASS 2024). We divide it into twelve 5-hectare patches, the scale at which grazing within a patch is approximately uniform (Barnes et al. 2008). The soil properties that initialize the biogeochemical model

are drawn from SSURGO (Soil Survey Staff 2026) for this location, and the net primary productivity of 319 grams of carbon per square meter per year is set by site calibration (Section 4.10).

Nominal mature-cow, replacement-heifer, and annual-calf targets use shares of 0.50, 0.10, and 0.40 with representative coefficients of 1.00, 0.70, and 0.20 LSU per head against a 500-kilogram reference animal. These values convert the nominal annual herd target H to a cow target. Realized opening class proportions emerge from the demographic controller, which initializes attrition-consistent cow and heifer counts around that target, holds calves as a pending cohort until calving, and thereafter supplies the same daily class counts to the animal-demand and accounting modules. Cows calve on day 75 and calves are weaned near 213 kilograms on day 270. At weaning, the existing heifer cohort is promoted to cows, surviving calves are split equally by sex, and the retention controller keeps enough female calves to form the next replacement-heifer cohort. Male and surplus female calves are sold, and any replacement shortfall is purchased. At the year boundary, cow and heifer mortality is 1.5 percent and the baseline cow culling rate is 12 percent; the resulting counts and next pending calf cohort initialize the next annual cycle. Cattle graze on pasture from day 120 to 270 and are fed stored forage off-pasture from October through April, following the Michigan grazing calendar.

Daily exogenous forcing drives the biophysical model. The shared 2025 start uses the gridMET forcing (Abatzoglou 2013) delivered with the continuous 1926–2045 MEMS history. From 2026 onward, the experiment replays gridMET years beginning in 1979, while every management cell receives the same calendar and atmospheric context. The comparison is therefore conditional on one common forcing path. We study thirteen nominal stocking rates from 0.8 to 3.2 LSU per hectare in increments of 0.2 and defer the full cross with paddock count and grazing period to Section 4.11. Restricting the demonstration to one site isolates management effects from between-site variation, while the surrogate makes the expanded grid computationally feasible.

4.2. Integrated computational architecture

Figure 1 organizes the components by time scale and makes their nesting explicit. Annual setup consists of the whole-farm management configuration and realized opening herd composition. On each day, the calendar and animal-demand schedule provide parallel inputs; animal selective grazing chooses the occupied patch; and local LSU per hectare and the zero supplement input drive the management-responsive biophysical transition. The corresponding MEMS process constructs removal, trampling, and animal-return fluxes internally. The resulting state is the input for the next day. At calving, weaning, and the year boundary, the herd-composition controller updates class counts and determines next year’s pending calf cohort. A single annual farm account combines the two nonneg-

ative feed streams, herd outcomes, SOC, and all revenue and cost items. The ten-year management comparison lies outside the one-year box and aggregates repeated annual cycles.

The Microbial Efficiency-Matrix Stabilization model (MEMS) supplies the process-based transition by simulating carbon and nitrogen cycling at daily resolution (Zhang et al. 2024; Santos et al. 2024). The management-responsive surrogate replaces that transition inside the farm loop and uses the preceding patch state, site and weather context, daily local LSU per hectare, and optional on-pasture supplement. All 663 management cells use the surrogate. Purchased winter and grazing-season feed enter the economic account, but none is applied to the biophysical transition, so $s_{k,t,d} = 0$ throughout the experiment.

4.3. MEMS process model and patch representation

MEMS is a process-based model of carbon and nitrogen cycling among soil layers, vegetation, and the atmosphere at daily resolution (Zhang et al. 2024; Santos et al. 2024). It resolves SOC into measurable pools whose turnover responds to weather, plant inputs, and grazing offtake. We treat the pool structure as given and cite the model description for the mechanism. For this paper, the relevant outputs are the daily aboveground forage state and the annual farm-level SOC stock.

We represent the 60-hectare farm as K biophysically identical patches, each simulated by its own MEMS instance sharing one soil profile, one weather sequence, and one forage species mix. The patches differ only in their dynamic state, that is, in the biomass and leaf-to-stem composition that grazing history produces. With a patch area of 5 hectares, the farm contains $K = 12$ patches. This spatial resolution corresponds to the scale at which grazing within a patch is treated as approximately uniform and is held fixed across all management comparisons.

4.4. Surrogate design, training, and validation protocol

The surrogate predicts the complete next end-of-day state from the preceding state, fixed site characteristics, target-day weather and calendar context, and daily management. MEMS’s unobserved internal compartment transfers remain latent. Deterministic accounting identities and physical output links preserve forage accounting, non-negativity, bounds, and internal consistency. Appendix C documents the complete input–output specification.

The 7.9-million-parameter management-responsive model combines a depth-aware encoder for the sixteen soil layers with an encoder for the non-layered state, site, weather, calendar, and management inputs. Its management vector is $\mathbf{a}_{k,t,d} = (u_{k,t,d}, s_{k,t,d})$: daily local LSU per hectare and optional on-pasture supplement. The fused representation

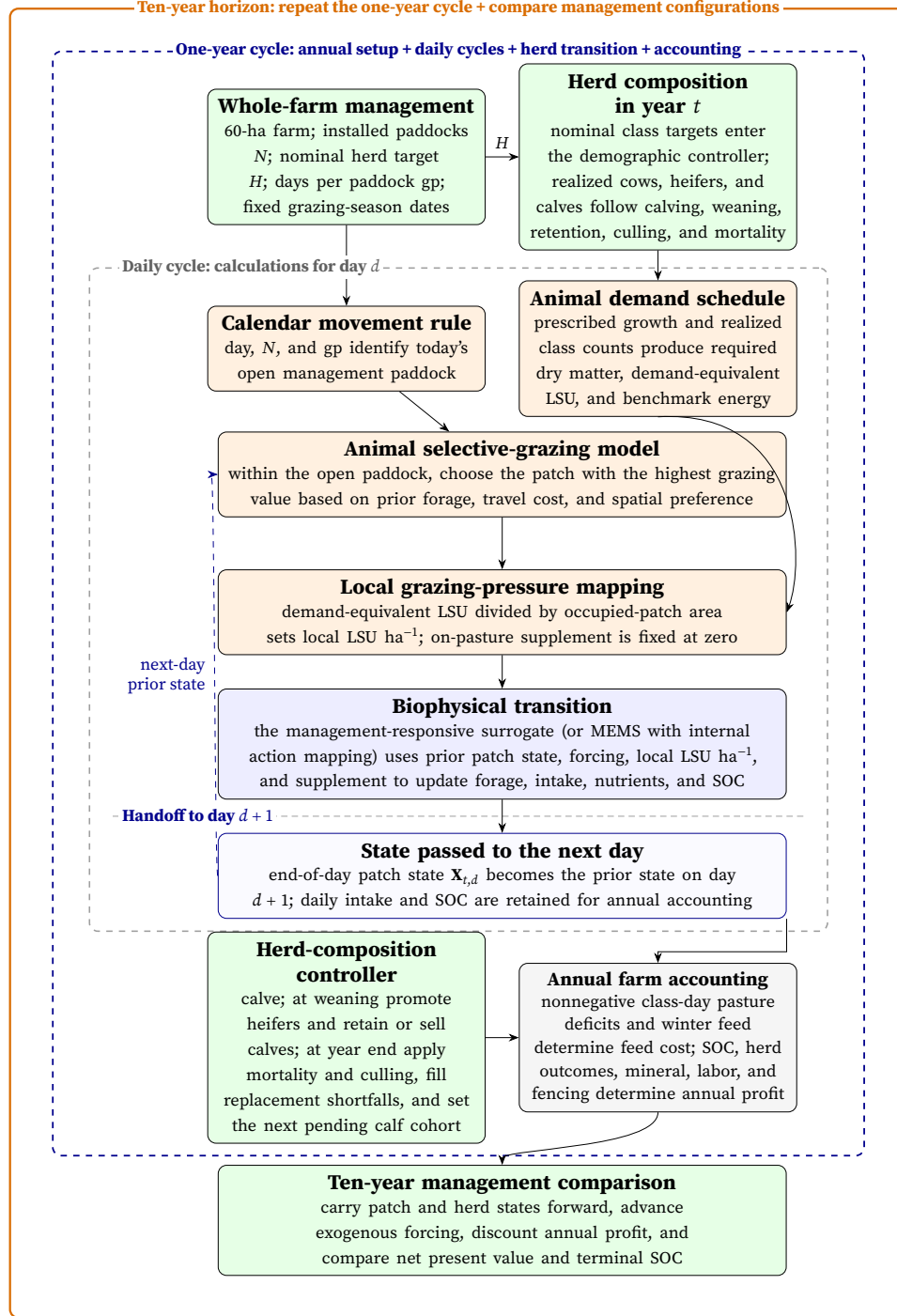


FIGURE 1. Temporal nesting and information flow. Annual setup establishes the whole-farm management configuration and realized herd composition. On day d , the calendar and animal demand determine the eligible area and demand-equivalent herd; animal selective grazing, local-pressure mapping, and the biophysical transition produce the state used on day $d + 1$. After the daily cycle has been repeated over the year, a one-way accounting handoff carries accumulated daily outcomes into the annual farm account. Annual accounting remains downstream of the daily transition. The herd controller and farm account close the year. The outer loop carries patch and herd states and exogenous forcing across ten annual cycles and compares discounted private profit and terminal SOC.

enters a two-layer gated recurrent unit (GRU) with 768 hidden features.

For patch k on day d , its recursion can be written compactly as

$$(19) \quad \begin{aligned} e_{k,t,d} &= \Phi_{\theta}(\widehat{X}_{k,t,d-1}, z_k, w_{t,d}, \mathbf{a}_{k,t,d}), \\ h_{k,t,d} &= \text{GRU}_{\theta}(e_{k,t,d}, h_{k,t,d-1}), \\ \widehat{X}_{k,t,d} &= \mathcal{R}_{\text{phys}}(\widehat{X}_{k,t,d-1}, \Psi_{\theta}(h_{k,t,d})). \end{aligned}$$

Here θ collects learned parameters, Φ_{θ} encodes the state and daily inputs into feature vector $e_{k,t,d}$, $h_{k,t,d}$ is the GRU memory carried between days, and Ψ_{θ} maps that memory to the output heads. The farm-management layer constructs $u_{k,t,d}$ causally before the transition, and $\mathcal{R}_{\text{phys}}$ reconstructs the next physical state while applying output links and constraints. Most output heads predict a residual change to the preceding state; soil temperature is predicted in levels. The GRU therefore gives the recursive emulator a learned memory of recent weather, vegetation, soil, and management history in addition to the explicit preceding state.

Training uses 320 approximately thirty-year MEMS trajectories, half of which are active-management runs that vary local grazing density, grazing and rest duration, timing, and management history. The training objective combines a transition-scaled smooth-L1 loss with penalties for recursive drift, the learned reserve-carbon gate, and persistent level error. WAPE serves a separate purpose as the validation and model-selection metric. Appendix C.2 gives the architecture, objective, and training details.

For farm simulation, the calendar and animal selective-grazing layers determine $u_{k,t,d}$ before the transition. The management-responsive approximation is

$$(20) \quad \begin{aligned} F_{\text{MEMS}}(X_{k,t,d-1}, z_k, w_{t,d}, \mathcal{A}(X_{k,t,d-1}, u_{k,t,d}, s_{k,t,d})) \\ \simeq \widehat{F}_{\theta}(X_{k,t,d-1}, z_k, w_{t,d}, u_{k,t,d}, s_{k,t,d}), \end{aligned}$$

where $\mathcal{A}$ denotes the internal action construction used by MEMS. Paddock count, farm geometry, calendar scheduling, and selective movement remain outside the neural model in the farm-management layer. The experiment sets $s = 0$. Appendix C.1 gives the complete management input-output specification.

Validation tests recursive prediction over the long horizons relevant to management analysis. The final architecture is evaluated on eight ten-year trajectories, and its performance from the management experiment's 2025 initial state is checked separately against a ten-year ungrazed MEMS continuation. Section 5 reports both comparisons; Appendix C.3 defines the metric and validation panel.

4.5. Calendar movement and selective patch use

The numerical implementation specifies the abstract grazing value in equation (6) as

$$(21) \quad \Omega_{k,t,d} = P_k [1 - \mathcal{T}(k_{t,d-1} \rightarrow k)] \mathcal{J}(B_{k,t,d-1}) Q_{k,t,d-1}^\gamma,$$

where P_k is a fixed spatial-preference weight, $\mathcal{T}$ is the travel-cost term, $\mathcal{J}(B)$ is a saturating intake response to biomass B , Q is live-leaf share, and γ governs the strength of quality selection. The score is read from the preceding end-of-day state, so selection is causal.

The two grazing regimes share identical patches, nominal herd target, and season length and differ only in how the calendar restricts the herd's eligible area. Under continuous grazing, all K patches remain eligible throughout the grazing season; this regime is represented by $N = 1$. Under calendar-based rotation, equation (A2) activates one of $N > 1$ management paddocks for g_p days. In both regimes, equations (21) and (6) implement animal selective grazing by choosing the patch with the highest grazing value within the eligible area. The value combines a saturating (Type II) intake response to biomass, live-leaf share, travel cost, and a fixed preference weight for non-forage attractors such as water. Thus the management regime determines where cattle are allowed to graze, while animal selective grazing determines where they graze within that eligible area. The contrast at fixed nominal herd target isolates the within-farm redistribution channel of Section 3.10. Figure 2 shows the patch layout and the two grazing regimes.

4.6. Herd nutrition and stored-feed accounting

The animal model maps prescribed liveweight trajectories into feed demand and evaluates the energy supplied by realized forage following National Research Council relationships (National Academies of Sciences, Engineering, and Medicine 2016). The herd has three classes—cows, replacement heifers, and calves—and uses the nominal shares and reference LSU coefficients specified in Section 4.1. The demographic controller determines realized class counts, which can differ from the nominal shares.

We prescribe each class a piecewise-linear weight trajectory over the year and invert the energy chain to recover the daily intake the trajectory requires. Cows calve at day of year 75, losing weight at parturition and then regaining it over the season, while calves grow from birth to a weaning weight near 213 kilograms at day of year 270, when they are sold. Running the energy chain in reverse, from a target daily gain to the dry-matter intake that supports it, lets the trajectory set forage demand directly and keeps each class on its calibrated growth path.

The prescribed trajectory and realized counts supply total required dry matter $D_{t,d}^{\text{req}}$. Equation (5) converts that requirement to the demand-equivalent herd $L_{t,d}$ used in equation (7); this daily input generally differs from the nominal annual target H . The same

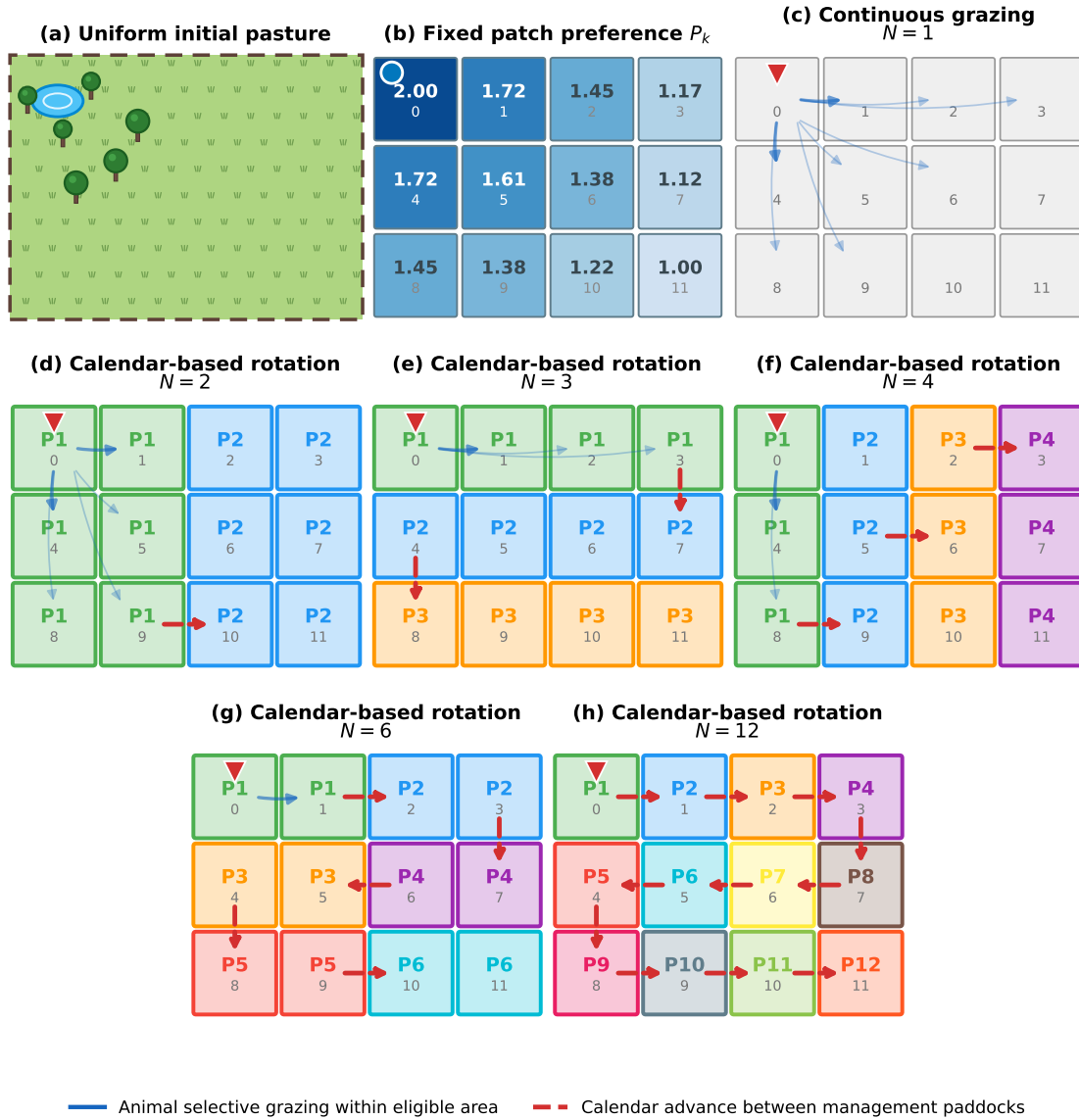


FIGURE 2. Continuous grazing, calendar-based rotation, and animal selective grazing. The 60-hectare farm contains $K = 12$ equal 5-hectare biophysical patches. Panel (a) shows their common initial pasture conditions and the fixed corner water-and-shade attractor; panel (b) shows the resulting patch-preference gradient P_k . Under continuous grazing (c), all patches remain eligible throughout the grazing season; this regime is represented by $N = 1$. Under calendar-based rotation (d)–(h), equation (A2) activates one of $N > 1$ management paddocks for gp days. In both regimes, equations (21) and (6) select the highest-value eligible biophysical patch. When $N = 12$, each management paddock contains one patch and the calendar fully determines location. Colors identify management-paddock membership, blue arrows show animal selective grazing within the eligible area, and red dashed arrows show calendar advances between management paddocks.

class-specific demand supplies the benchmark energy in equation (B1), valued at 10 percent crude protein. The surrogate then advances the grazing season with $s_{k,t,d} = 0$ and reports realized forage carbon and nitrogen intake. The animal module uses their ratio to infer realized crude protein, distributes realized intake among classes in proportion to body-size-weighted demand, and computes $E_{c,t,d}^{\text{past}}$ in equation (B2). Equation (10) converts every positive class-day deficit to hay-equivalent dry matter. Surplus for one class or day cannot offset a deficit elsewhere.

Off-season hay S_t^{off} is calculated separately because the herd consumes no pasture then. Winter feed and grazing-season deficit feed are independently nonnegative and priced separately before being added in equation (11). These charges support the prescribed growth curve through the economic account alone; forage, intake, manure, SOC, and animal weight remain governed by the daily simulation. The model also handles calving, weaning, culling, and replacement accounts. Appendix Figure B1 shows the prescribed trajectories.

4.7. Daily coupled simulation

The coupled simulation follows the daily ordering in Figure 1. At the start of target day d , the demand schedule evaluates that day's class counts from the cohort sizes and fixed event calendar, then computes $D_{t,d}^{\text{req}}$ and $L_{t,d}$. The calendar and preceding patch state determine location and local management inputs, the management-responsive surrogate advances every patch, and the resulting state passes to day $d + 1$. Current-day animal demand and patch selection therefore enter the same daily transition. On-pasture supplement is zero, and the stored-feed account is computed only after the pasture simulation. Every cell begins from the same state on day of year 119 in 2025, taken from a continuous MEMS simulation of ungrazed history spanning 1926–2025. Appendix B.2 records the complete execution order and the extension reserved for future positive-supplement scenarios.

4.8. Economic accounting

Profit is calculated downstream of the biophysical and herd simulation. Within a model run, grazing, SOC, and herd dynamics are therefore unaffected by the accounting result. On a cash basis, annual profit is revenue minus cost:

$$(22) \quad \pi_N(y) = \underbrace{R_{\text{calf}}(y) + R_{\text{cull}}(y)}_{\text{revenue}} - \underbrace{\left[C_{\text{feed}}(y) + C_{\text{herd}}(y) + C_{\text{min}}(y) + C_{\text{labor}}(y) + C_{\text{fence}}(y) + C_{\text{repl}}(y) \right]}_{\text{cost}}.$$

Each term is evaluated for simulation year y . Revenue comes from calf sales R_{calf} and cull-cow sales R_{cull} , priced per hundredweight. Feed cost C_{feed} separately prices winter feed and the grazing-season deficit feed in equation (11); the broad ERS purchased-, homegrown-, and grazed-feed aggregates are excluded to avoid double counting. The remaining costs are herd nonfeed operations C_{herd} , mineral C_{min} , labor C_{labor} (a per-cow base plus scheduled-move labor), the annual equivalent of perimeter and interior fencing C_{fence} , and replacement-heifer purchases C_{repl} . For a T -year management scenario beginning in year y_0 , the reported net present value is

$$(23) \quad \text{NPV}_N^{(T)} = \sum_{\tau=0}^{T-1} \delta^\tau \pi_N(y_0 + \tau), \quad \delta = (1 + 0.05)^{-1}.$$

We set $T = 10$ and $y_0 = 2025$, so the first management year has discount factor one. Consistent with the cash basis, the private-profit objective carries no carbon revenue. Terminal SOC is reported as a separate environmental outcome, while the theoretical shadow value λ of Section 3.9 measures SOC's private continuation value through forage and feed-cost savings.

4.9. Numerical assumptions and scope

Five assumptions are especially important for interpreting the numerical results. First, nominal herd target, paddock count, grazing period, and season dates remain constant within each fixed management cell; dynamic optimization is left for future work. Second, the nominal target H represents intended annual herd size, while realized class counts and prescribed dry-matter demand determine the daily demand-equivalent input $L_{t,d}$. Third, animal growth and sale weights follow prescribed annual trajectories and remain invariant to realized forage. Fourth, every positive class-day pasture-energy deficit incurs a grazing-season feed charge. Deficits are additive across classes and days and are also separate from winter feed. Both purchased-feed streams remain in the economic account, and the on-pasture supplement input is zero. Fifth, every cell receives the same deterministic exogenous forcing path. Appendix B.3 states the full set of assumptions.

4.10. Parameterization

Appendix Table B1 summarizes the parameters used in the numerical experiment, and a machine-readable registry accompanying the simulation output provides the complete specification. Site calibration and the movement-rule calibration are documented separately in the supplementary technical documentation. The biological and technical parameters fix the 60-hectare, twelve-patch Michigan farm, the day-of-year 120–270 grazing season, the three-class herd trajectories and demographic rules, and the calendar and

animal selective-grazing rules described above. These quantities are held constant across management cells; only nominal stocking rate, installed paddock count, and grazing-period length vary.

Calves and cull cows are valued at \$264 and \$115 per hundredweight, respectively, with a 500-kilogram cull weight. Winter feed is priced at \$220 per short ton of consumed dry matter, a rounded hay-equivalent value derived from Michigan hay prices using 85 percent dry matter and 90 percent recovery. Grazing-season deficit feed is priced separately at the same \$220 hay-equivalent rate. Annual nonfeed herd cost is \$280.84 per cow and mineral cost is \$54.75 per cow. Labor combines a \$27.99 per-cow base, a \$19.70 hourly wage, and move time of 30 minutes plus 0.10 minute per cow over a 24-week labor season. Permanent fence costs \$2.51 per foot with a 20-year life, temporary fence costs \$1.05 per foot with a 10-year life, and replacement heifers cost \$1,200 per head. We discount at 5 percent. Calf and cull prices are fixed calibration values, and the replacement-heifer price is treated as a model assumption.

4.11. Management experiment design

We evaluate every cell in a finite management grid. Nominal stocking rate ranges from 0.8 to 3.2 LSU per hectare in 0.2-unit increments, equivalent to 48–192 nominal LSU on the 60-hectare farm. At each of the 13 stocking rates, continuous grazing is represented by $N = 1$, with all patches eligible throughout the grazing season. Calendar-based rotation uses $N \in \{2, 3, 4\}$ with $gp \in \{1, 2, 3, 4, 5, 6, 7, 10, 15, 20, 25, 30\}$ days, and $N \in \{6, 12\}$ with $gp \in \{1, 2, 3, 4, 5, 6, 7\}$ days. Rest remains the derived quantity $(N - 1)gp$. The rotation schedule is fixed within a season, but the starting management paddock advances across years; within the active paddock, equation (21) supplies the same animal selective-grazing rule used under continuous grazing. Thus each stocking rate has one continuous-grazing configuration and $3(12) + 2(7) = 50$ calendar-based-rotation configurations, for 51 management configurations per stocking rate and $13(51) = 663$ management cells in total.

Every cell starts on day of year 119 in 2025 from the same state generated by a continuous MEMS simulation of ungrazed history for 1926–2025. The farm and financial accounts cover ten management years, 2025–2034; SOC change compares day of year 119 in 2025 with the same date in 2035. Weather for 2025 comes from the historical MEMS simulation, and the experiment replays gridMET weather beginning with 1979 from 2026 onward. All cells share this forcing and simulation context, so their differences isolate management under that path. The management-responsive surrogate advances each patch state from two daily inputs: local grazing density and optional on-pasture supplement dry matter. The experiment sets the on-pasture supplement input to zero. For farm accounting, total realized pasture intake is divided among animal classes in proportion to their metabolic body size, $n_{c,t,d} w_{c,t,d}^{0.75}$. The model then purchases enough grazing-season feed to cover

each class's positive daily energy shortfall relative to its prescribed growth path; a forage surplus for one class or day cannot offset a deficit for another. Appendix B.1 defines this calculation formally.

The design yields 663 scenario summaries and 6,630 annual financial–SOC observations. All simulations completed successfully. These scenarios hold management fixed, so the reported optimum is the highest-NPV configuration in the tested grid; solving the Bellman problem lies beyond the present experiment. The surrogate's speed makes this expanded comparison practical and provides a basis for later mixed discrete–continuous optimization.

5. Surrogate validation and computational performance

We evaluate the surrogate's recursive agreement with MEMS, its performance from the initial state used in the management experiment, and its computational speed. Section 6 then reports the 663 management scenarios generated with the surrogate.

5.1. Recursive predictive fidelity

Figure 3 combines a representative management-relevant recursive rollout with validation across eight fixed trajectories. Panels (a)–(c) compare MEMS and surrogate predictions for one representative biophysical-patch trajectory, while panel (d) reports average accuracy across the validation set. Appendix Table C1 gives the channel averages and defines the metric. The arithmetic mean of the 64 trajectory–channel WAPes is 6.44 percent. Mean WAPE is 8.31 percent for forage quantity, 11.20 percent for vegetation composition, and 0.44 percent for SOC stocks.

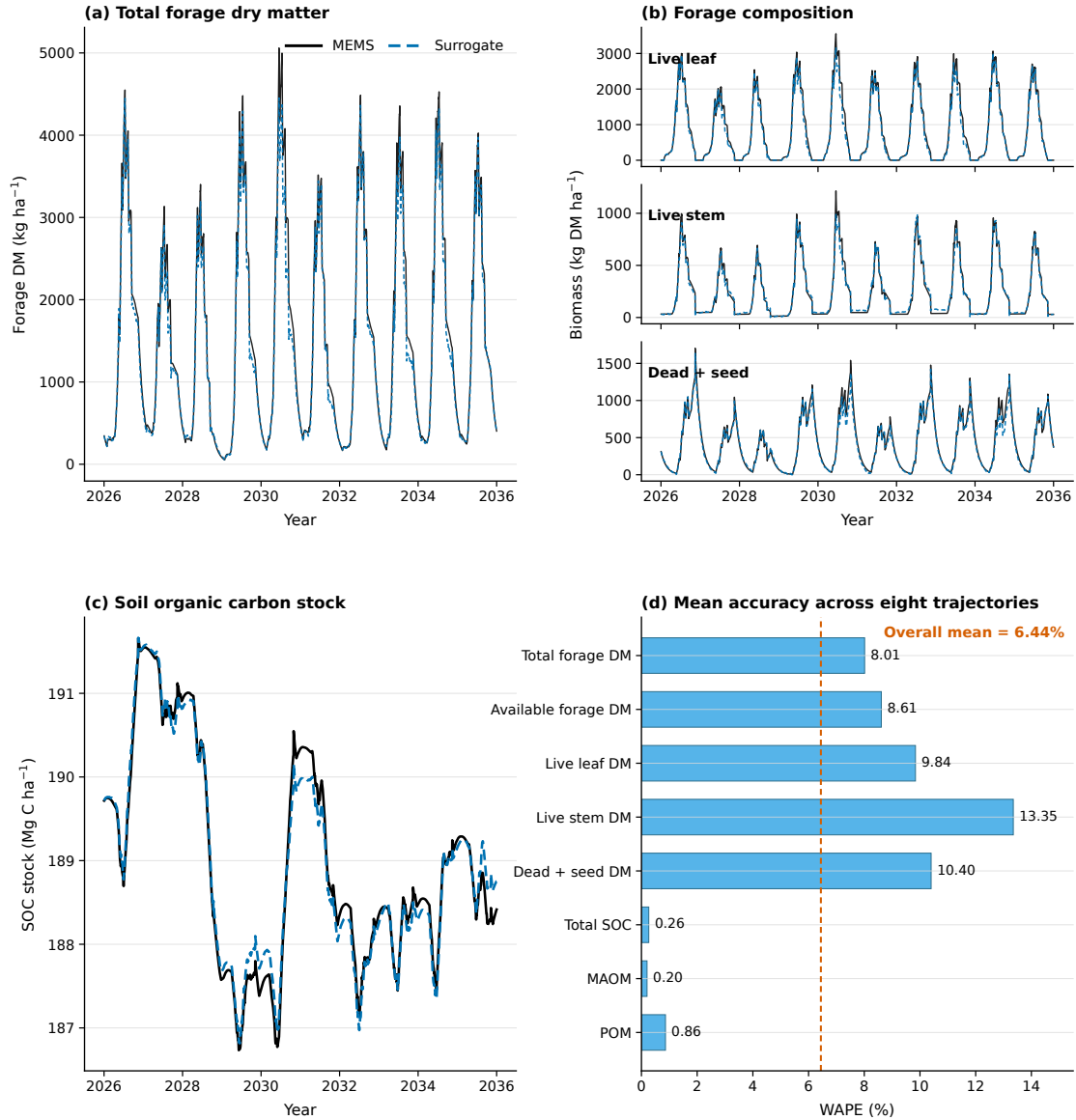


FIGURE 3. Ten-year recursive agreement between MEMS and the final surrogate. Panels (a)–(c) compare daily MEMS outputs (solid black) with recursive surrogate predictions (blue dashed) for a representative 5-hectare biophysical-patch trajectory from a calendar-based-rotation cell with nominal stocking rate $SR = 1.0 \text{ LSU ha}^{-1}$, $N = 4$ management paddocks, and grazing-period length $gp = 7$ days. The panels show total forage dry matter, its live-leaf, live-stem, and dead-plus-seed components, and the total-profile SOC stock. The surrogate is initialized from the true state once and subsequently advances from its own previous prediction while recorded forcing and management are replayed. Panel (d) reports the mean ten-year level WAPE for each management-relevant output across the eight fixed validation trajectories, computed from their daily values; the vertical dashed line marks the overall mean of 6.44 percent across the 64 trajectory–output combinations. Panels (a)–(c) illustrate timing and long-horizon drift for one management-relevant trajectory, while panel (d) summarizes accuracy across the complete validation panel.

These results establish stable average agreement with MEMS for the outcomes used to compare management. Vegetation composition has a higher mean WAPE than aggregate forage quantity and SOC stocks, with live-stem dry matter having the largest channel-average error at 13.35 percent. All eight validation rollouts complete with zero non-finite predictions, negative protected states, or soil-temperature-envelope violations.

5.2. Farm integration and computational speed

The management-responsive surrogate is connected to the calendar scheduler, selective movement rule, prescribed herd, nutrition calculation, and economic accounting module. Its daily inputs are local LSU per hectare and optional on-pasture supplement, as in equation (20); the 663-scenario experiment sets supplement to zero. Starting from the 2025 initial state used in the management experiment, a ten-year ungrazed comparison with MEMS has a largest output-specific WAPE of 7.31 percent, while total SOC and MAOM WAPE are 0.23 and 0.10 percent. This comparison verifies performance under the initial state and forcing used in the management scenarios. Appendix C.1 records the interface.

A ten-year adaptive-management scenario takes approximately one hour with iterative MEMS and approximately 20 seconds with surrogate inference, an approximately 181-fold reduction in model runtime. This comparison concerns model execution time only.

The surrogate generates all 663 management scenarios in Section 6. The outcomes inherit the process representation and limitations of MEMS and await independent field validation. Positive on-pasture supplementation is outside the reported experiment because the supplement input is fixed at zero.

6. Management-scenario results

The 663 scenarios characterize grid-wide private-profit and SOC trade-offs (Section 6.1), identify when and why calendar-based rotation pays relative to continuous grazing (Section 6.2), and show how calendar-based rotation redistributes grazing across biophysical patches (Section 6.3). They are surrogate predictions for a transient decade at one site under one forcing path, initialized from the common 2025 pre-grazing state. Modeled SOC declines in every cell relative to the baseline generated by an ungrazed MEMS history from 1926 to 2025. The carbon results therefore support a relative-loss comparison within this model configuration; they provide no verified sequestration estimate.

6.1. Private-profit and soil-carbon trade-offs

Figure 4 summarizes private NPV and SOC outcomes across all 663 cells. In panels (a) and (b), the shaded envelope spans the lowest and highest outcome among the calendar-based-rotation configurations at each stocking rate. The shading displays variation across

designed management configurations; statistical uncertainty lies outside the experiment. The dashed black line is continuous grazing, while the colored line follows the calendar-based rotation with the highest private NPV at each stocking rate. Both envelopes widen as stocking pressure rises, indicating that paddock count and calendar timing matter most when forage demand is high. Across the full grid, ten-year NPV ranges from a loss of \$15,112 to a gain of \$200,046. The least SOC loss is -8.15 Mg C per hectare at 0.8 LSU per hectare with 12 paddocks and a six-day grazing period. The largest loss is -58.24 Mg C per hectare at 3.2 LSU per hectare with 12 paddocks grazed one day at a time; this cell also has the lowest NPV.

Panel (c) shows that the private-profit and SOC-loss rankings diverge. The grid-wide private-NPV maximum uses 2.0 LSU per hectare, two paddocks, and a 20-day grazing period. Its ten-year NPV is \$200,046 and its modeled SOC change is -32.35 Mg C per hectare. The second-ranked management cell uses 2.2 LSU per hectare, two paddocks, and a 20-day grazing period (NPV \$197,826), and the third uses 2.0 LSU per hectare, four paddocks, and a seven-day grazing period (NPV \$197,403). The maximum is the only cell within 1 percent of the highest NPV. It is the best tested grid point; a continuous optimum remains unestimated.

The least SOC loss occurs at 0.8 LSU per hectare, 12 paddocks, and a six-day grazing period, with an NPV of \$72,871 and SOC change of -8.15 Mg C per hectare. The dark outlines in panel (c) identify 37 nondominated cells between that least-loss endpoint and the private-NPV maximum: no other tested management cell has NPV at least as high and SOC loss no greater, with a strict improvement in one outcome. Because the private objective assigns no carbon revenue, the frontier expresses two criteria in their original economic and physical units.

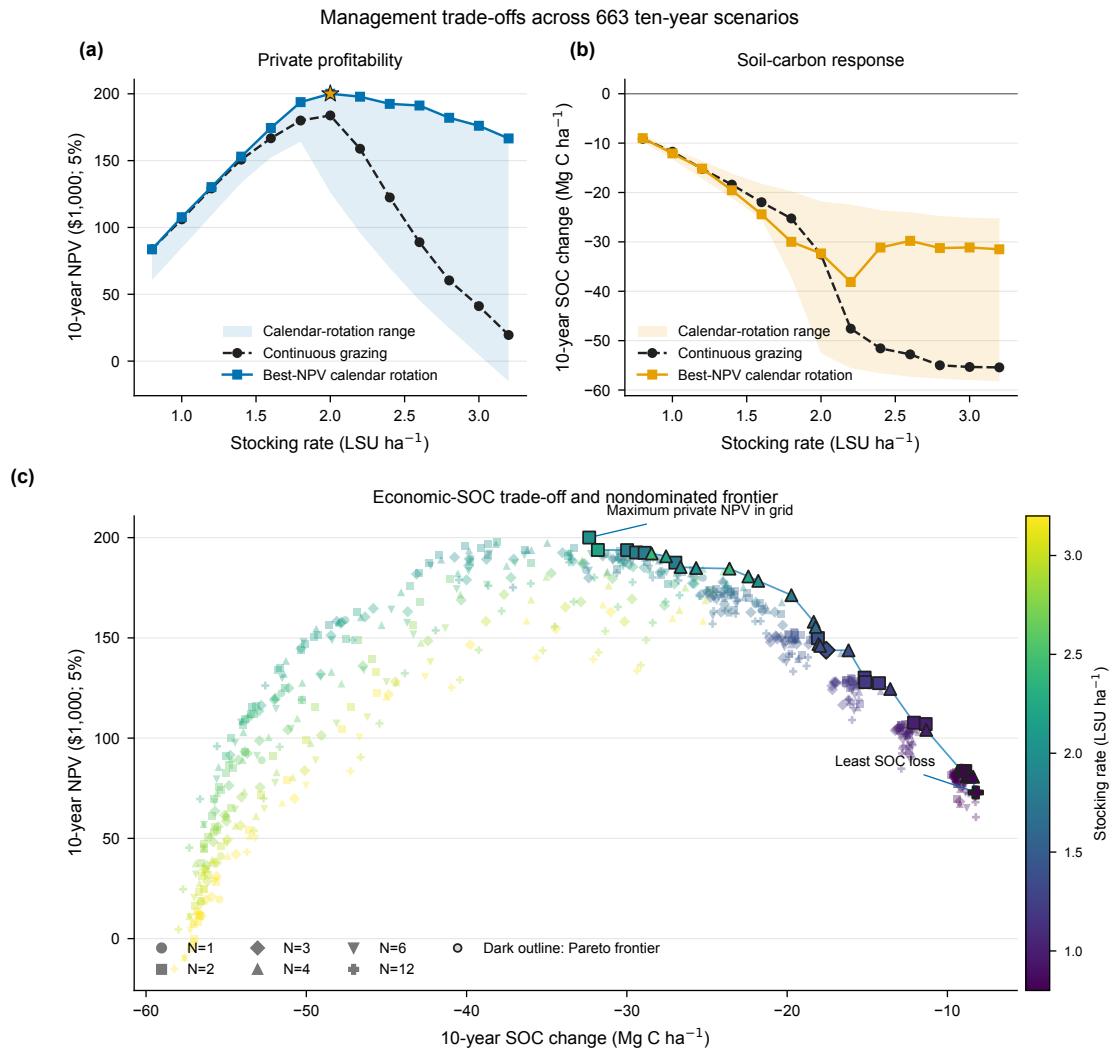


FIGURE 4. Private-NPV and soil-carbon trade-offs across 663 management scenarios. Panels (a) and (b) plot ten-year private NPV and total-profile SOC change against nominal stocking rate. Shading spans the range across calendar-based-rotation configurations ($N > 1$) at each stocking rate and summarizes management variation; the figure contains no statistical uncertainty band. The dashed black line is continuous grazing, represented by $N = 1$, and the colored square line is the calendar-based rotation with the highest private NPV at that stocking rate; the star marks the maximum private NPV in the tested grid. Panel (c) plots every management cell in economic–SOC space. Symbols identify installed paddock count, color identifies stocking rate, and dark outlines identify the 37 nondominated cells. Outcomes farther upward and to the right are preferred.

6.2. When and why calendar-based rotation pays

The comparison between the highest-NPV calendar-based rotation and continuous grazing changes sign between 0.8 and 1.0 LSU per hectare. At 0.8 LSU per hectare, the highest-NPV calendar-based rotation trails continuous grazing by 0.13 percent. From 1.0 LSU per hectare upward, at least one calendar-based-rotation configuration pays, although the percentage advantage is not monotonic at low stocking. The advantage is 1.51 percent at 1.0, 0.77 percent at 1.2, 8.85 percent at 2.0, and 753.06 percent at 3.2 LSU per hectare, where continuous-grazing NPV falls to \$19,531 as the grazing-season deficit-feed bill rises. Appendix Table D1 gives the complete stocking-rate comparison.

The feed account explains this stocking-rate dependence. At low stocking, reductions in the grazing-season deficit-feed charge are small relative to the fencing and move labor required by calendar-based rotation. At high stocking, the highest-NPV calendar-based rotations sharply reduce that charge relative to continuous grazing. The charge converts positive pasture-energy deficits into a hay-equivalent feed quantity that reconciles the economic account with prescribed growth. It remains outside the pasture simulation and is never passed to the surrogate. Within a stocking rate, every management configuration uses the same prescribed weight paths, realized herd outcomes, and 213-kilogram calf sale weight, so revenue is identical. Configuration-level profit differences arise from grazing-season deficit feed, move labor, and fencing; animal growth is fixed across configurations.

Figure 5 traces annual outcomes and ten-year financial composition for five exact scenarios. At 2.0 LSU per hectare, continuous grazing returns a ten-year NPV of \$183,788, whereas calendar-based rotation with two paddocks and a 20-day grazing period returns \$200,046, a gain of \$16,259 or 8.85 percent. Calendar-based rotation lowers the grazing-season deficit-feed cost by \$21,885 over ten years, more than offsetting \$1,048 of additional labor and \$2,211 of fencing. The resulting undiscounted cost reduction is \$18,626; the difference between that amount and the NPV gain reflects annual discounting. Calendar-based rotation also slightly moderates modeled SOC loss: it ends at 189.50 Mg C per hectare, compared with 189.30 under continuous grazing, equivalent to losses of 32.35 and 32.55 Mg C per hectare from their common baseline.

The high-stocking contrast is more pronounced. At 3.2 LSU per hectare, calendar-based rotation with four paddocks and a 20-day grazing period raises NPV by \$147,078 and avoids 23.92 Mg C per hectare of SOC loss relative to continuous grazing. The continuous-grazing cell earns \$17,981 in the first management interval but loses money in six of the next nine; annual profit under calendar-based rotation remains positive in all ten intervals, ranging from \$13,843 to \$31,402. The ten-year undiscounted cost reduction is \$186,887, driven by \$193,404 less grazing-season deficit-feed cost and partly offset by \$1,181 more labor and \$5,337 more fencing. The highest-NPV calendar-based rotation first improves both NPV and SOC relative to continuous grazing at 1.2 LSU per hectare and does so consistently from

2.0 LSU per hectare upward, so the joint gain remains conditional on stocking pressure.

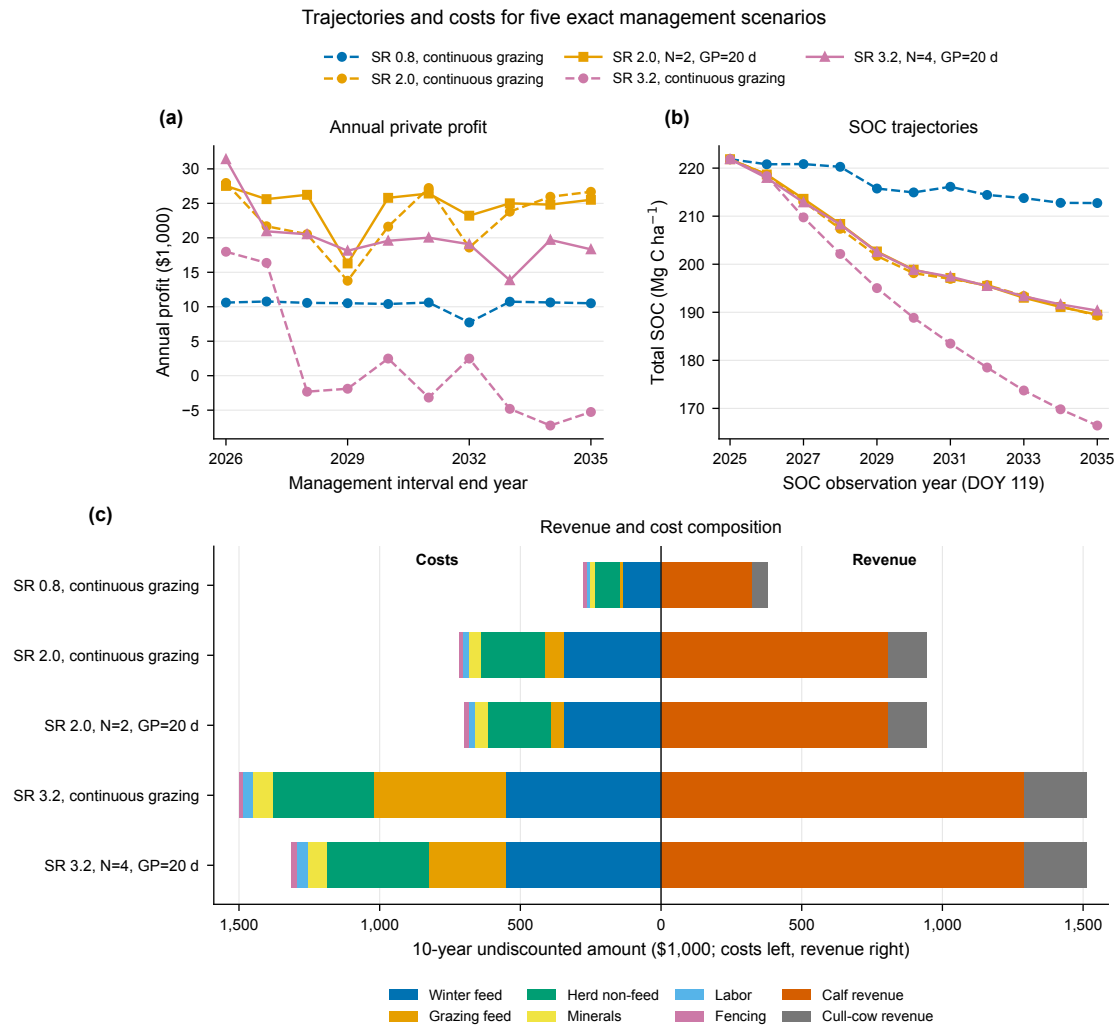


FIGURE 5. Annual profit, SOC trajectories, and ten-year financial composition for five management scenarios. The scenarios are continuous grazing at 0.8 LSU ha⁻¹, paired continuous-grazing and highest-NPV calendar-based-rotation configurations at 2.0 LSU ha⁻¹, and the corresponding pair at 3.2 LSU ha⁻¹. Panel (a) reports annual private cash profit, with each year identifying the end of its management interval. Panel (b) reports total-profile SOC at day of year 119, beginning from the common 2025 state. Panel (c) decomposes undiscounted ten-year costs (left) and revenues (right); the bars show cumulative cash flows before discounting. Within each stocking-rate pair, prescribed animal production and revenue are identical, so the financial difference comes from feed and management costs. Continuous grazing is represented by $N = 1$; calendar-based rotation uses $N > 1$.

6.3. How calendar-based rotation redistributes patch use

Figure 6 quantifies one spatial pathway through which calendar-based rotation can influence management outcomes: the redistribution of patch use. On each grazing-season day, the whole herd occupies one 5-hectare biophysical patch, so every ten-year scenario allocates 1,510 patch-occupancy days across the same 12 patches. Patch-use imbalance is the largest patch count divided by the smallest. A ratio of one means that all patches are occupied equally often. The statistic counts days of patch occupancy and contains no information about cattle-days or within-patch grazed area.

Under continuous grazing, the median imbalance across the 13 stocking rates is 3.09 and the interquartile range is 2.31–24.75. Calendar restriction generally reduces that concentration, most strongly when the installed design approaches one management paddock per biophysical patch. For 12 paddocks and grazing periods of one to seven days, median imbalance ranges only from 1.02 to 1.11. Intermediate designs remain sensitive to timing: at 30-day grazing periods, the medians rise to 2.85 for three paddocks and 2.74 for four paddocks. Spatial uniformity therefore depends on the combination of paddock count and grazing period, which governs how long animal selective grazing operates within the eligible area.

Six individual cells have an infinite imbalance ratio because at least one patch is never occupied: the continuous-grazing cell at 1.0 LSU ha^{-1} and five three-paddock calendar-based-rotation cells at 0.8 LSU ha^{-1} . The medians and interquartile ranges plotted in Figure 6 remain finite despite these cases. Selective underuse thus persists in some management cells despite calendar restriction. Because the outcomes jointly reflect forage, SOC, animal, and cost channels, the figure cannot isolate the redistribution effect. It is also a model result awaiting field validation of cattle locations.

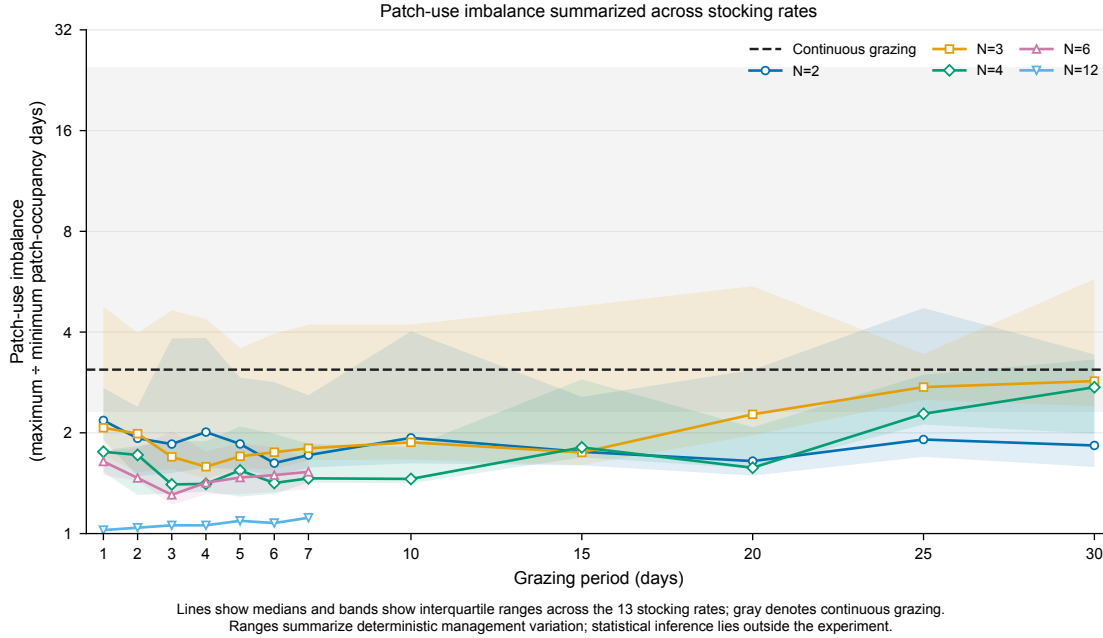


FIGURE 6. Patch-use imbalance under continuous grazing and calendar-based rotation. The outcome is the ratio of the maximum to minimum ten-year patch-occupancy counts; one denotes equal use. Lines report medians and bands report interquartile ranges across the 13 stocking rates for each paddock-count–grazing-period treatment under calendar-based rotation. These ranges summarize variation across the deterministic management grid; no statistical uncertainty is estimated. The dashed line and gray band are the continuous-grazing median and interquartile range. The vertical axis uses a base-2 logarithmic scale. Six individual cells have infinite ratios because their minimum count is zero, although every plotted median and interquartile range is finite. Under calendar-based rotation, the calendar determines the active management paddock; under both regimes, the animal selective-grazing rule determines the occupied biophysical patch within the eligible area.

7. Conclusions and discussion

7.1. Main conclusions and contributions

SOC enters the private grazing problem as productive natural capital because its effects on forage quantity and quality alter stored-feed requirements and farm profit. The associated shadow value is the discounted private return to an additional unit of SOC. In the stationary theoretical system, accounting for future SOC changes affects the management choice: holding the nominal herd target fixed, a forward-looking producer selects a longer grazing period than a myopic producer when the marginal SOC effect remains positive at the myopic choice. The numerical comparison shows that calendar-based rotation pays only for particular combinations of stocking pressure, installed paddock count, and grazing-

period length. Continuous grazing leads at 0.8 LSU per hectare, whereas at least one calendar-based rotation leads from 1.0 LSU per hectare upward. The highest private NPV is \$200,046 at 2.0 LSU per hectare, two paddocks, and a 20-day grazing period, 8.85 percent above continuous grazing at the same stocking rate. Private-profit and SOC rankings also differ. Modeled SOC declines in every cell relative to the baseline generated by an ungrazed MEMS history from 1926 to 2025. The least loss occurs at 0.8 LSU per hectare, 12 paddocks, and a six-day grazing period, while the private-profit maximum occurs elsewhere.

The spatial and financial results identify redistribution as one contributor to these rankings. Calendar-based rotation restricts the area eligible for grazing and thereby limits selective concentration: median patch-use imbalance is 3.09 under continuous grazing, compared with 1.02–1.11 across the tested 12-paddock calendars. Six low-stocking cells still leave at least one patch unused, but the redistribution changes local grazing pressure and subsequent forage availability. Those changes operate within the larger system of forage and SOC dynamics, animal demand and growth, herd accounting, and management costs. At low stocking, any feed-cost savings are too small to offset fencing and movement costs. At higher stocking, selected calendar-based rotations reduce the grazing-season deficit-feed charge enough to generate a private gain; at 2.0 LSU per hectare, the lower feed bill net of fencing and movement costs accounts for the 8.85-percent NPV advantage. The profit and carbon performance of calendar-based rotation are conditional on stocking pressure, paddock design, timing, and the spatial forage state.

Beyond the Michigan estimates, the paper provides a tractable bridge between soil-process modeling and farm economics. It carries SOC as productive capital into a grazing problem that includes paddock design, herd and timing choices, daily selective movement, herd transitions, nutrition, and annual accounting. The recurrent surrogate reproduces the daily process transition with a mean WAPE of 6.44 percent across eight management-relevant outputs over eight ten-year recursive trajectories and no physical-validity violations. Reducing a ten-year adaptive-management run from approximately one hour with iterative MEMS to 20 seconds makes the 663-scenario comparison feasible while keeping management, movement, herd dynamics, nutrition, and economics outside the neural model. The patch representation adds a spatial contribution by measuring grazing redistribution as one mechanism connecting infrastructure and timing to forage use, feed costs, profit, and SOC.

7.2. Economic and policy interpretation

The management implication is to replace the binary question of whether “rotation works” with a conditional comparison of stocking rate, installed paddock count, and grazing-period length. Scheduled rest follows automatically from the calendar once managers choose the number of paddocks and how long the herd remains in each one. Adding

paddocks can redistribute grazing, but its private value depends on the combined effects on forage availability and deficit-feed cost and on whether any feed-cost savings exceed fencing and movement costs. The privately optimal calendar-based rotation can also differ from the configuration with the least SOC loss. The 37 nondominated cells describe this tension in economic and physical units: they identify management configurations for which no other tested cell has both at least as much private NPV and no more SOC loss.

Ecosystem-service incentives could select a different point from this frontier, although the payment required remains outside the present experiment. An outcome-based carbon payment would add value to measured SOC retention, a practice-based cost-share would reduce the private cost of paddock infrastructure or movement, and a qualifying-beef premium would operate through livestock revenue. Because these instruments enter different parts of the economic account, they can produce different management responses. Comparing them requires credible SOC response functions across sites and initial conditions, program costs, and a computed shadow value under a stationary reference environment. We provide the shadow-value expression and integrated framework needed for such comparisons; empirical ranking of the instruments remains future work.

7.3. Limitations and research priorities

The numerical findings are model-based scenario comparisons and have yet to be validated as field treatment effects. They come from one Michigan site, one deterministic forcing path, one common 2025 initial state, prescribed animal growth, and a discrete set of fixed ten-year management configurations. The highest-NPV result is therefore the best tested configuration; the annual Bellman problem remains unsolved. The simulated decade also falls short of the stationary reference environment used to define the theoretical shadow value. Consequently, the paper defines the Michigan steady-state shadow value of SOC and leaves its computation for future work. The rankings express private value and exclude the social value of carbon.

Two implementation limits are especially important. First, purchased winter feed and the class-specific grazing-season deficit-feed charge appear only as economic reconciliation accounts. On-pasture intake, manure, forage, and SOC remain unaffected, and the surrogate's on-pasture supplement input is zero. Ongoing work relaxes this assumption by allowing supplemental feed to enter both the daily biophysical transition and the economic account through a consistent placement rule. Second, under calendar-based rotation, the calendar determines advancement between management paddocks, while under both grazing regimes a deterministic animal selective-grazing rule chooses one 5-hectare patch from the eligible area using previous-day forage conditions. The reported patch-use measure therefore counts only how many days each patch is selected as the herd's location. It omits the number of animals using that patch, the area actually grazed

within it, behavioral differences among cattle, and forage-triggered movement between management paddocks; those movements follow the fixed calendar.

The next step is to evaluate whether the qualitative rankings persist across sites, initial SOC states, climate scenarios, policy settings, and market conditions. Multi-site experiments with alternative weather paths and locally appropriate calibration can reveal the heterogeneity relevant to ecosystem-service design (Stanley et al. 2026); variation in livestock and feed prices, labor, and fencing costs can test these economic channels. Simulating alternative land-use histories would supply a range of initial SOC states and permit calculation of the steady-state shadow value of SOC for comparison with cropland estimates (Berazneva et al. 2019). A mixed discrete–continuous optimizer can then replace the fixed grid and jointly search over nominal herd target, grazing-period length, and installed paddock design. Separate extensions should evaluate forage-triggered management-paddock advancement and positive on-pasture supplementation; the latter requires a daily placement controller and validation of supplement-sensitive intake and forage-quality pathways. Together, these extensions would show how robust the present profit–SOC trade-off is to the climatic, policy, and market environments in which grazing decisions are made.

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Appendix A. Theory derivations

This appendix supplies the bookkeeping and envelope steps behind the calendar results in Section 3.

A.1. Calendar identity and local grazing input

Under the equal-duration calendar, a complete cycle contains N consecutive grazing periods of gp_t days, so its length is $c_t = Ngp_t$. A management paddock is active for one such period and is inactive while the other $N - 1$ paddocks are active. Its scheduled rest is therefore

$$(A1) \quad r_t = c_t - gp_t = (N - 1)gp_t,$$

which gives equation (4). The identity records calendar bookkeeping. Biomass and residual forage respond to the schedule, while advancement occurs solely according to the calendar.

Let $j_{t,0}$ be the first management paddock used in year t . During the grazing season, the exact scheduler is

$$(A2) \quad j_{t,d} = 1 + \text{mod} \left(j_{t,0} - 1 + \left\lfloor \frac{d - d_{\text{on}}}{gp_t} \right\rfloor, N \right).$$

If the grazing season ends partway through a cycle, the final realized interval can be truncated; rotating the starting paddock across years distributes that boundary effect while preserving the nominal identity.

Management-paddock rest also differs from the biological rest of an individual patch. Equation (A2) makes every patch in the active management paddock eligible, but equation (6) occupies only the patch with the highest grazing value. An unselected patch can therefore remain ungrazed even while its management paddock is active. Conversely, all patches outside the active paddock receive zero local LSU by construction.

The local-input mapping preserves the whole-farm herd while concentrating it spatially. Because exactly one patch is occupied during the grazing season, equation (7) implies

$$(A3) \quad \sum_{k=1}^K \frac{A_k}{A} u_{k,t,d} = \frac{L_{t,d}}{A}.$$

The right-hand side is the demand-equivalent whole-farm LSU per hectare, whereas the occupied patch receives $L_{t,d}/A_k$. Thus the grazing regime determines the eligible set and its timing, animal selective grazing chooses the occupied patch, and the farm-management

layer passes the resulting daily local LSU per hectare to MEMS or the surrogate. When an on-pasture supplementation rule is active, the layer also passes the colocated input in equation (B4); that input is zero in the reported experiment.

A.2. Management benchmarks

Let $Y_N(\mathbf{X}_t, \mathbf{n}_t, H_t, \text{gp}_t)$ be pasture forage supplied to the herd during the year and let $\Delta\text{SOC}_N = g_N - \text{SOC}_t$ be annual SOC change. Conditional on the opening state, H_t , and N , the forage and SOC benchmarks are

$$\text{gp}_F^* \in \arg \max_{\text{gp}} Y_N, \quad \text{gp}_{\text{SOC}}^* \in \arg \max_{\text{gp}} \Delta\text{SOC}_N.$$

The myopic producer chooses

$$(H_\pi^*, \text{gp}_\pi^*) \in \arg \max_{H, \text{gp}} \pi_N(\mathbf{X}_t, \mathbf{n}_t, H, \text{gp}),$$

whereas the forward-looking producer solves equation (1). These definitions separate biological and current-profit benchmarks from the dynamic choice and leave their ordering unrestricted.

A.3. Envelope derivation of the shadow value

The shadow value (17) follows from the envelope theorem applied to the reduced scalar, stationary-reference-forcing version of equation (1) defined in Section 3.9. Write S for SOC, and let $H^*(S)$ and $\text{gp}^*(S)$ denote the maximizing controls conditional on installed N ; subscripts on π_N and g_N below denote partial derivatives. Differentiating $\tilde{V}_N(S) = \pi_N(S, H^*, \text{gp}^*) + \delta \tilde{V}_N(g_N(S, H^*, \text{gp}^*))$ gives

$$\begin{aligned} \tilde{V}'_N(S) &= \pi_{N,S} + \delta \tilde{V}'_N(g_N) g_{N,S} \\ (A4) \quad &+ [\pi_{N,H} + \delta \tilde{V}'_N(g_N) g_{N,H}] \frac{dH^*}{dS} + [\pi_{N,\text{gp}} + \delta \tilde{V}'_N(g_N) g_{N,\text{gp}}] \frac{d\text{gp}^*}{dS}. \end{aligned}$$

The two bracketed terms are the operating first-order conditions (15) and (16), so both vanish at an interior optimum. The envelope condition is therefore

$$(A5) \quad \tilde{V}'_N(S) = \pi_{N,S} + \delta \tilde{V}'_N(g_N) g_{N,S}.$$

Writing $\lambda_t = \tilde{V}'_N(\text{SOC}_t)$ and imposing a stationary scalar equilibrium gives $\lambda_N^{ss} = \pi_{N,S} + \delta \lambda_N^{ss} g_{N,S}$. Solving yields equation (17). The value is positive and finite when $\pi_{N,S} > 0$ and $\delta g_{N,S} < 1$: SOC has a positive marginal effect on current profit through avoided stored-feed purchases, but its discounted persistence is bounded.

A.4. Stationary operating system

Conditional on installed N , the stationary system has four unknowns, $(H^*, gp^*, SOC^{ss}, \lambda_N^{ss})$, and four equations:

$$\begin{aligned}
 (SS-1) \quad & \frac{\partial \pi_N}{\partial H} + \delta \lambda_N^{ss} \frac{\partial g_N}{\partial H} = 0, \\
 (SS-2) \quad & \frac{\partial \pi_N}{\partial gp} + \delta \lambda_N^{ss} \frac{\partial g_N}{\partial gp} = 0, \\
 (SS-3) \quad & g_N(SOC^{ss}, H^*, gp^*) = SOC^{ss}, \\
 (SS-4) \quad & \lambda_N^{ss} = \frac{\partial \pi_N / \partial SOC}{1 - \delta \partial g_N / \partial SOC}.
 \end{aligned}$$

Here H^* is the stationary nominal herd target. The first two equations determine the operating margins, the third is the SOC fixed point, and the fourth is the envelope-based shadow value. The discrete installed-paddock designs are compared directly by value and have no additional first-order condition.

A.5. Conditional dynamic extension of the grazing period

Fix the current state, installed design N , and nominal herd target H . Let $gp_m = gp^m(H; N)$ be an interior maximizer of current profit, so $\pi_{N, gp}(S, H, gp_m) = 0$. The derivative of the dynamic objective with respect to the grazing period, evaluated at that myopic choice, is

$$(A6) \quad \left. \frac{\partial}{\partial gp} [\pi_N(S, H, gp) + \delta \tilde{V}_N(g_N(S, H, gp))] \right|_{gp=gp_m} = \delta \lambda_{t+1} g_{N, gp}(S, H, gp_m).$$

If $\lambda_{t+1} > 0$ and $g_{N, gp} > 0$ at gp_m , this derivative is positive. If the dynamic objective is strictly concave in gp over the relevant interval, its conditional maximizer lies to the right: $gp^d(H; N) > gp^m(H; N)$. Equation (A1) then gives $r^d(H; N) > r^m(H; N)$ for $N > 1$. This result holds conditional on H and N . When herd target and timing change together, their interaction is governed by the full system (15)–(16), which leaves the coordinate-by-coordinate ordering unrestricted.

Appendix B. Numerical implementation and calibration

This appendix gives the accounting identities, execution order, assumptions, and parameter values underlying the numerical model in Section 4.

B.1. Energy-based stored-feed accounting

Let $n_{c,t,d}$ be the number of animals in class c , let $D_{c,t,d}^{\text{req}}$ be prescribed dry-matter intake per head, let $\bar{q}$ be the reference crude-protein fraction, and let $e_m(q)$ be usable maintenance energy per kilogram of dry matter at crude-protein fraction q . The class-level daily energy benchmark implied by the prescribed growth path is

$$(B1) \quad E_{c,t,d}^{\text{bench}} = e_m(\bar{q})n_{c,t,d}D_{c,t,d}^{\text{req}}.$$

The numerical model sets $\bar{q} = 0.10$. Let $I_{t,d}^{\text{past}}$ be realized total pasture dry-matter intake and let $q_{t,d}$ be the crude-protein fraction inferred from the carbon and nitrogen in forage consumed from the selected patch. If $w_{c,t,d}$ is representative liveweight per head, class c receives the metabolic-body-size share

$$\omega_{c,t,d} = \frac{n_{c,t,d}w_{c,t,d}^{0.75}}{\sum_{c' \in \mathbb{C}} n_{c',t,d}w_{c',t,d}^{0.75}}, \quad \sum_c \omega_{c,t,d} = 1.$$

Class-level pasture-supplied energy is

$$(B2) \quad E_{c,t,d}^{\text{past}} = e_m(q_{t,d})\omega_{c,t,d}I_{t,d}^{\text{past}}.$$

Both quantities are measured in megacalories per day. The benchmark is fixed by the herd and prescribed growth path, whereas pasture-supplied energy varies with realized forage intake and quality. With $e_s = e_m(q_s)$ denoting usable energy per kilogram of hay-equivalent dry matter, the implementation applies the positive-part operator before aggregation:

$$(B3) \quad S_t^g = \sum_{d=d_{\text{on}}}^{d_{\text{off}}} \sum_{c \in \mathbb{C}} \frac{\max\{0, E_{c,t,d}^{\text{bench}} - E_{c,t,d}^{\text{past}}\}}{e_s}.$$

This is equation (10). It prevents pasture-energy surplus for one class or day from offsetting a deficit elsewhere. Positive off-season hay is calculated independently and added through equation (11); offsets between the two streams are disallowed.

For completeness, let $a_{t,d}^{\text{past}} \geq 0$ denote an explicitly applied on-pasture supplement in kilograms of dry matter. If a future placement rule colocates it with the herd, the corresponding local input is

$$(B4) \quad s_{k,t,d} = \mathbf{1}\{k = k_{t,d}\} \frac{a_{t,d}^{\text{past}}}{10A_k},$$

in grams of dry matter per square meter per day when A_k is measured in hectares. The reported baseline sets $a_{t,d}^{\text{past}} = 0$, so $s_{k,t,d} = 0$.

B.2. Coupled execution

The patch states advance one day at a time through the grazing season, day of year 120 to 270. At the start of target day d , the animal layer evaluates $n_{c,t,d}$ from the cohort sizes and event calendar and computes $D_{t,d}^{\text{req}}$ and the same-day demand-equivalent herd $L_{t,d}$. The grazing regime determines the eligible area: all patches under continuous grazing or the active management paddock under calendar-based rotation. The animal selective-grazing rule reads the end-of-day- $d-1$ patch states and selects the occupied patch, and equation (7) assigns that patch the day's local grazing pressure. The management-responsive surrogate then advances every patch through day d from local LSU per hectare and the zero supplement input. The resulting state becomes the input to day $d+1$. Outside the grazing season every patch advances with zero grazing. Class-day grazing deficits and off-season hay are calculated after the biophysical simulation and priced as separate nonnegative feed streams.

In a future positive-supplement scenario, the animal layer would first calculate the day- d shortfall from information available at the start of the day. A placement rule would then convert the realized amount to $s_{k,t,d}$, and the same amount would enter both the surrogate and the economic account before the state advances. That extension requires validation of supplement-sensitive intake and forage-quality outputs. The present analysis keeps $s_{k,t,d} = 0$.

Every simulation begins from the same day-of-year-119 state in 2025, extracted from a continuous MEMS simulation of ungrazed history for 1926–2025. The surrogate then carries that state recursively through the management decade.

B.3. Numerical assumptions and scope

The numerical implementation rests on the following assumptions:

1. *Fixed management grid.* Each management cell holds its nominal herd target, installed paddock count, and grazing period constant across all years. Grazing-season dates are fixed. The analysis compares these scenarios; dynamic optimization under equations (SS-1)–(SS-4) remains future work.
2. *Nominal and daily LSU are distinct.* The design label H/A uses the reference class coefficients in equation (2); daily local pressure instead uses demand-equivalent $L_{t,d}$ from equation (5).
3. *Consistent demographic accounting.* Pending calves before calving and live calves afterward belong to the same cohort, and the demographic controller's realized cow, heifer, and calf counts drive both daily demand and annual accounting.
4. *Identical patches.* The twelve patches share one soil, weather sequence, and species

mix. Their within-farm heterogeneity emerges from divergent grazing histories after a common initialization.

5. *Deterministic movement.* The herd follows equation (21) deterministically, so each management cell yields a single trajectory.
6. *Prescribed, year-invariant growth.* Each herd class follows a fixed within-year weight trajectory that repeats annually. Prescribed gains remain identical across weather and forage-quality realizations.
7. *One representative weight per class.* All animals in a class share one weight; there is no within-class weight distribution.
8. *Economic stored-feed accounting.* Every positive class-day pasture-energy deficit is converted to hay-equivalent dry matter. Deficits are additive across classes and days and remain separate from positive winter feed. Purchased feed stays in the economic account, and on-pasture supplement is zero.
9. *Exogenous forcing.* Every management cell receives the same forcing path described in Section 4.11: historical weather is replayed, whereas atmospheric carbon dioxide follows the simulation year.
10. *No economic feedback within a simulation.* Profit is computed after the simulation and never alters biophysical or herd dynamics. The cash basis excludes land rent, the opportunity cost of grazed forage, and carbon payments.
11. *Calendar management-paddock advancement.* The calendar alone determines when the herd advances to the next management paddock. Within the active paddock, previous-day biomass and forage quality affect the daily grazing value and can change which biophysical patch is selected through equation (21). A forage-triggered advancement rule is reserved for future analysis.

B.4. Calibration and parameter values

Table B1 reports the parameters needed to interpret and reproduce the management comparison. A machine-readable registry distributed with the simulation output contains the complete sweep, movement, initialization, surrogate, animal-nutrition, prescribed-growth, and economic specification. Supplementary technical documentation provides additional site-calibration information and movement-rule provenance. Appendix C houses the surrogate architecture and validation settings.

Site and farm. The study site is in Michigan. The 60-hectare farm is a representative-farm design assumption for the numerical experiment. The 5-hectare patch size sets the resolution at which grazing within a patch is approximately uniform, dividing the farm into twelve patches. Net primary productivity and the initializing soil state are calibrated

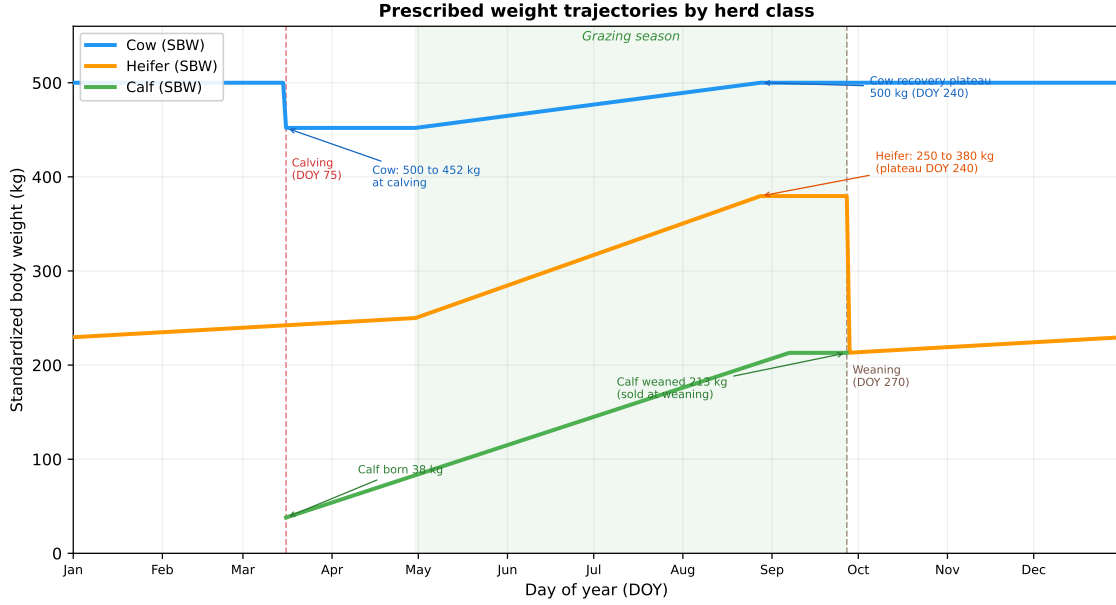


FIGURE B1. Prescribed growth curves by herd class. Cows lose weight at calving (day of year 75) and recover over the grazing season, and heifers and calves gain through the season to their weaning and replacement targets. Inverting the energy chain from these trajectories yields the daily forage demand the herd places on the pasture.

to the site.

Herd. Nominal class shares and per-head reference livestock-unit coefficients come from the USDA Census class structure together with herd-survey composition, against a 500-kilogram reference animal. They convert H to a nominal cow target; the demographic controller then constructs attrition-consistent opening counts and event-driven daily counts. For example, $H = 60$ gives the nominal target vector (46.15, 9.23, 36.92) for cows, heifers, and calves, but the first opening state contains 40.01 cows, 6.24 heifers, no live calves, and 40.01 pending calves that enter on the calving date. Cows follow a piecewise prescribed trajectory: standardized body weight is 500 kilograms before calving, falls to 452 kilograms after calving, and recovers from day 120 to 500 kilograms by day 240. Heifers enter the grazing season on day 120 at 250 kilograms and gain 1.08 kilograms per day to day 240; calves enter at birth on day 75 at 38 kilograms and gain 1.00 kilogram per day to day 250, yielding the 213-kilogram sale weight used at weaning on day 270. Thus only the heifer and calf paths use constant daily-gain parameters; cow gain is implied by the piecewise trajectory. Calving and weaning dates and the demographic rates in Table B1 are fixed across cells. The demand-equivalent conversion uses the MEMS daily intake capacity in equation (5).

Movement rule. The half-saturation biomass, handling time, and quality exponent of the movement rule (21) are set from the functional-response and bite-mechanics literature

on grazing intake and from the quality-preference literature. The travel-cost coefficients and the nine-hour daily grazing time budget come from grazing-behavior studies and movement-rule calibration, and the preference map encodes a corner-water gradient as the non-forage attractor.

Economics. Calf and cull prices are fixed calibration values, and the replacement-heifer price is treated as a model assumption. Physical winter and grazing-season feed replace the broad ERS feed aggregates, which are excluded to prevent double counting; winter feed is derived from Michigan hay prices, while grazing-season deficit feed is a separate hay-equivalent parameter set to the same price. Herd nonfeed operations, mineral, labor, fence, and replacement costs remain separate. The discount rate is 5 percent per year, and profit is computed on a cash basis with no carbon revenue, consistent with equation (22).

TABLE B1. Selected parameter values and model settings for the 663-cell numerical experiment.

Quantity	Value	Unit	Source / note
Site and farm			
Farm area	60	ha	representative-farm design assumption
Patch area, count	5, 12	ha, —	within-patch uniformity scale
Net primary productivity	319	$\text{g C m}^{-2} \text{ yr}^{-1}$	site calibration
Soil texture (sand, clay)	48–91, 8–23	%	SSURGO, 0–150 cm profile
Soil pH	5.8–7.3	—	SSURGO
Grazing season	120 to 270	day of year	Michigan off-pasture October to April
Herd composition			
Nominal class shares (cow / heifer / calf)	0.50 / 0.10 / 0.40	—	management-target conversion; USDA Census + herd survey
LSU coefficients	1.00 / 0.70 / 0.20	—	500 kg reference weight
Nominal weighted LSU per head $\bar{\ell}$	0.65	—	derived from target shares
MEMS intake ratio $\rho_D \chi_{\text{MEMS}}$	0.0258	d^{-1}	0.03×0.86 demand reconciliation
Demand capacity κ_{MEMS}	12.9	$\text{kg DM LSU}^{-1} \text{ d}^{-1}$	equation (5)
Calving, weaning	75, 270	day of year	fixed herd calendar
Cow weight (pre-calving / post-calving / peak)	500 / 452 / 500	kg head^{-1}	piecewise prescribed trajectory

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Quantity	Value	Unit	Source / note
Cow recovery plateau	240	day of year	no constant cow-gain parameter
Heifer weight (grazing start / off-season reset)	250 / 213	kg head ⁻¹	prescribed-growth calibration
Heifer gain; plateau	1.08; 240	kg d ⁻¹ ; day of year	extension benchmark + calibration
Calf weight (birth / weaning)	38 / 213	kg head ⁻¹	prescribed-growth calibration
Calf gain; plateau	1.00; 250	kg d ⁻¹ ; day of year	external benchmark + calibration
Mortality (cow / heifer / calf)	0.015 / 0.015 / 0.030	fraction yr ⁻¹	demographic controller
Cow culling rate	0.12	fraction yr ⁻¹	demographic controller
Female-calf ratio; heifer retention rate	0.50; 0.15	fraction	demographic controller
Movement rule (equation 21)			
Half-saturation biomass $B_{1/2}$	1,200	kg DM ha ⁻¹	intake-response calibration
Handling time h	1.5	s g ⁻¹ DM	bite-mechanics literature
Quality exponent γ	2	—	quality-preference literature
Travel cost (α_0, α)	0.25, 0.02	—, per 100 m	movement-rule calibration
Grazing time budget	9	h d ⁻¹	grazing-behavior literature
Preference map P_k	corner-water gradient	—	non-forage attractor (water)
Economics (cash basis, equation 22)			
Calf price	264	\$ cwt ⁻¹	fixed calibration value
Cull-cow price	115	\$ cwt ⁻¹	fixed calibration value
Cull-cow sale weight	500	kg head ⁻¹	model assumption
Winter feed price	220	\$ short ton ⁻¹ consumed DM	Michigan hay price, 85% DM, 90% recovery; rounded
Grazing-season deficit-feed price	220	\$ short ton ⁻¹ consumed DM	separate hay-equivalent parameter; 12% CP
Herd nonfeed cost	280.84	\$ cow ⁻¹ yr ⁻¹	broad ERS feed aggregates excluded
Mineral cost	54.75	\$ cow ⁻¹ yr ⁻¹	separate annual account
Labor base, wage	27.99, 19.70	\$ cow ⁻¹ , \$ h ⁻¹	farm labor parameterization
Move time	30 + 0.10 n_{cow}	min move ⁻¹	24-week grazing season

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Quantity	Value	Unit	Source / note
Fence (permanent, temporary)	2.51 / 20, 1.05 / 10	\$ ft ⁻¹ / yr life	Missouri Extension costs; NRCS/product-life assumptions
Replacement heifer	1,200	\$ head ⁻¹	model assumption
Stored-feed crude protein q_s	0.12	fraction of DM	winter and grazing-season deficit-feed account
Discount rate	5	% yr ⁻¹	analysis convention
Design sweep			
Nominal stocking rate H/A	0.8–3.2 by 0.2	LSU ha ⁻¹	48–192 nominal LSU on 60 ha
Paddock count N	1, 2, 3, 4, 6, 12	—	continuous grazing ($N = 1$); calendar-based rotation ($N > 1$)
Grazing period, $N = 2, 3, 4$	1–7, 10, 15, 20, 25, 30	d	12 levels
Grazing period, $N = 6, 12$	1–7	d	7 levels
Scheduled management-paddock rest	$(N - 1)$ gp	d	calendar identity
Rotation start	advances by year	management paddock	fixed rotation; balances incomplete seasonal cycles
Common initial state	ungrazed 1926–2025	MEMS history	day-0 state is 2025 day 119
Forcing path	2025 history; 2026+ re-play from 1979	gridMET year	identical path across cells
Management years; SOC endpoints	2025–2034; 2025–2035	calendar yr	SOC observed on day 119
Reported SOC pools	total / MAOM / POM / residual	Mg C ha ⁻¹	consistent annual reporting definition
Biophysical transition engine	management-responsive recurrent surrogate	—	advances every patch state one day at a time
Surrogate management inputs	local grazing density; on-pasture supplement	LSU ha ⁻¹ ; g DM m ⁻² d ⁻¹	supplement input set to zero in every cell
Grazing-season feed shortfall	feed needed to cover each class's positive daily energy gap	kg DM	purchased feed; no offset across classes or days
Class share of total pasture intake	animal count × body weight ^{0.75} , normalized across classes	fraction	heavier animals receive more per head; shares sum to one
Management cells	663	—	6,630 annual observations

Appendix C. Surrogate architecture and validation

This appendix reports the architecture, training objective, training data, and validation details for the final surrogate described in Section 4.4.

C.1. Input–output and management specification

The recurrent physical state contains 425 channels from nine MEMS output families: soil organic matter, crop, soil nutrients, soil temperature, soil water content, surface organic matter, surface feces, standing dead biomass, and water-storage states. Layered variables retain the sixteen-layer soil profile. Each daily transition also receives 177 fixed site inputs and 14 target-day weather and calendar inputs. The neural model predicts the complete next daily state; forage availability and verified algebraic identities are reconstructed deterministically from the predicted physical channels.

The final surrogate receives two target-day management inputs: daily local LSU per hectare and on-pasture supplement dry matter. It learns the direct mapping on the right side of equation (20); MEMS constructs the corresponding removal, trampling, fecal, and urinary fluxes internally on the left side. Paddock count, geometry, calendar scheduling, and selective movement stay in the farm-management layer. The reported experiment uses zero on-pasture supplement.

C.2. Architecture, training objective, and training data

The final structured recurrent model has 7.9 million parameters. A depth-convolution branch with three residual depth blocks encodes the sixteen-layer state and site profiles, while a two-layer multilayer perceptron encodes non-layered state, site, weather, calendar, and management inputs. A fusion layer combines the two representations before the two-layer, 768-feature GRU in equation (19). Per-channel heads map the GRU state to daily outputs. Most heads produce residual changes that are added to the preceding physical state, while the soil-temperature head predicts an absolute level. Physical output links enforce non-negativity and bounded variables, a soil-temperature envelope constrains the temperature profile, and the forage-availability projector and verified identities remain deterministic.

The training loss and WAPE serve different purposes. Let $\mathcal{D}$ be the set of days in a training window and let $x_{j,d}$ and $\hat{x}_{j,d}$ be the target and prediction for channel j on day d in that output head’s native representation—a daily change for residual heads and a level for absolute heads. Let $\rho_1(v)$ denote the smooth-L1 function with unit threshold, let m_j identify learned channels, let σ_j be the transition scale for a residual head or the level scale for an absolute head, and let w_j be the channel weight. The principal state-transition

loss is

$$(C1) \quad \mathcal{L}_{\text{trans}} = \frac{1}{\sum_j w_j m_j} \sum_j w_j m_j \left[\frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \rho_1 \left(\frac{\hat{x}_{j,d} - x_{j,d}}{\sigma_j} \right) \right].$$

The weights are updated from recursive-rollout normalized mean absolute error: channels with larger rollout error receive more weight, with weights clipped to the interval $[0.1, 10]$. This aligns allocation of training effort with long-horizon predictive performance, while WAPE remains reserved for validation.

Three auxiliary terms address failure modes beyond the one-day transition loss. The drift term $\mathcal{L}_{\text{drift}}$ penalizes the absolute signed mean level error over a training window on residual-state channels, preventing small daily biases from accumulating recursively. The gate term $\mathcal{L}_{\text{gate}}$ is binary cross-entropy for whether reserve carbon is drawn. Finally, for the seven directly predicted decision channels $\mathcal{J}_7$ —total forage, live leaf, live stem, dead plus seed, total SOC, MAOM, and POM—the pushed-rollout level term is

$$(C2) \quad \mathcal{L}_{\text{level}} = \frac{1}{7|\mathcal{D}|} \sum_{j \in \mathcal{J}_7} \sum_{d \in \mathcal{D}} \rho_1 \left(\frac{\hat{y}_{j,d} - y_{j,d}}{|y_{j,d}| + 0.15 \overline{|y_j|}} \right),$$

where $y_{j,d}$ and $\hat{y}_{j,d}$ are the true and predicted physical levels and $\overline{|y_j|}$ is channel j 's mean absolute training level. This term makes persistent errors visible even during low-biomass or nearly stationary periods. The resulting objectives are

$$(C3) \quad \mathcal{L}^{\text{teacher}} = \mathcal{L}_{\text{trans}} + 0.30\mathcal{L}_{\text{drift}} + 0.05\mathcal{L}_{\text{gate}}, \quad \mathcal{L}^{\text{push}} = \mathcal{L}^{\text{teacher}} + 0.20\mathcal{L}_{\text{level}}.$$

Teacher-forced windows train against true preceding states; pushed windows feed the surrogate's own preceding prediction back through equation (19). In the final training schedule, the horizon curriculum increases from one day to 1,095 days while the pushed-window probability rises to 0.90. Optimization uses AdamW with learning rate 6×10^{-5} , weight decay 10^{-4} , a 500-step warmup, and gradient-norm clipping at 1.0.

Training uses 320 approximately thirty-year MEMS trajectories. The first 160 cover the original site-scenario design. The other 160 are active-management trajectories with local grazing densities from 1 to 32 LSU per hectare, grazing periods from 1 to 14 days, rest periods from 7 to 60 days, varied first-graze timing, and multiple inherited management histories. These trajectories expose the model to repeated grazing-rest transitions and concentrated local grazing pressure while preserving the same daily state, site, context, and management-input specification used in the farm simulations.

C.3. Recursive validation protocol and detailed results

The validation panel contains eight trajectories selected to span sites and management regimes. Each rollout begins from the true initial state and then predicts recursively for 3,652 days, using its own previous prediction at the next step while replaying recorded weather and management. The eight decision channels are total forage dry matter, available forage dry matter, live leaf dry matter, live stem dry matter, dead plus seed dry matter, total profile SOC, profile MAOM, and profile POM.

For trajectory r and decision channel j , level WAPE is

$$(C4) \quad \text{WAPE}_{rj} = 100 \frac{\sum_d |\hat{y}_{rjd} - y_{rjd}|}{\sum_d |y_{rjd}|}.$$

The reported channel mean averages equation (C4) across the eight trajectories. Overall mean WAPE is the arithmetic mean of all 64 trajectory–channel WAPEs. These recursive rollouts evaluate stability across the sites and management regimes represented in the validation panel.

TABLE C1. Average validation performance of the final surrogate over eight ten-year recursive trajectories.

Decision channel	Mean WAPE (%)
Total forage dry matter	8.01
Available forage dry matter	8.61
Live leaf dry matter	9.84
Live stem dry matter	13.35
Dead plus seed dry matter	10.40
Total profile SOC	0.26
Profile MAOM	0.20
Profile POM	0.86
Overall	6.44

All eight ten-year rollouts complete with zero non-finite outputs, negative protected states, or soil-temperature-envelope violations. In the additional ungrazed comparison from the common 2025 initial state, the largest channel WAPE is 7.31 percent, while total SOC and MAOM WAPE are 0.23 and 0.10 percent. This validated management-responsive surrogate generates the 663-scenario experiment reported in Section 6.

Appendix D. Detailed management-sweep results

Table D1 compares continuous grazing with the highest-NPV calendar-based rotation at every stocking rate. “CG” denotes continuous grazing. The calendar-based rotation is reported by paddock count N and grazing period gp in days. The difference column measures that rotation relative to continuous grazing at the same stocking rate and is negative when continuous grazing has the higher NPV.

TABLE D1. Continuous-grazing and highest-NPV calendar-based-rotation outcomes by nominal stocking rate in the 663-cell surrogate experiment.

SR (LSU ha ⁻¹)	CG NPV (\$)	Best rotation	Rotation NPV (\$)	Difference (%)	CG ΔSOC (Mg C ha ⁻¹)	Rotation ΔSOC
0.8	83,751	$N = 2, gp=20$	83,639	-0.13	-9.12	-9.00
1.0	106,094	$N = 2, gp=20$	107,698	1.51	-11.74	-12.08
1.2	129,204	$N = 2, gp=20$	130,192	0.77	-15.22	-15.17
1.4	150,742	$N = 2, gp=7$	153,000	1.50	-18.41	-19.55
1.6	166,734	$N = 2, gp=4$	174,441	4.62	-21.97	-24.41
1.8	179,993	$N = 2, gp=4$	193,775	7.66	-25.26	-29.99
2.0	183,788	$N = 2, gp=20$	200,046	8.85	-32.55	-32.35
2.2	158,882	$N = 2, gp=20$	197,826	24.51	-47.57	-38.13
2.4	122,423	$N = 4, gp=15$	192,464	57.21	-51.56	-31.15
2.6	88,975	$N = 4, gp=20$	191,226	114.92	-52.78	-29.79
2.8	60,384	$N = 4, gp=20$	182,059	201.50	-54.99	-31.25
3.0	41,140	$N = 4, gp=20$	176,078	328.00	-55.33	-31.13
3.2	19,531	$N = 4, gp=20$	166,609	753.06	-55.42	-31.50